

Epidemiology of Kidney Disease

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KEY POINTS

- The definition of chronic kidney disease (CKD) has remained stable since 2002, with staging now including cause, glomerular filtration rate, and albuminuria categories.
- The prevalence of CKD globally is approximately 10%, with a suggestion of higher and rising prevalence in low- and middle-income countries.
- Risk factors for CKD incidence and progression provide a foundation for prevention with risk equations providing quantitative summaries, which can guide risk-based prevention.
- Genetic susceptibility to CKD is now better understood including a large effect by the *APOL1* susceptibility locus in populations of African descent.
- The incidence of end-stage kidney disease has stabilized at a high level in the United States and numerous other high-income countries, but enormous racial and ethnic disparities exist globally, with rates increasing in low- and middle-income countries.
- Kidney transplantation provides the best treatment for end-stage kidney disease. Despite a limited supply of organs, the prevalence of transplant recipients is increasing markedly, forming an important subset of patients with CKD.
- Acute kidney injury is both a consequence of CKD and a cause of further CKD progression.

There has been increased interest in the epidemiology of kidney disease in recent decades, fueled by the high and rising prevalence of kidney failure, growing recognition that chronic kidney disease (CKD) is common with a global prevalence of approximately 10%, and observations of disproportionate disease burden by region, race, and clinical characteristics. Kidney disease epidemiology, the study of disease incidence, distribution, and determinants, has enabled substantial progress in the field. Notable developments over the past decade include the issuance of guidelines for acute kidney disease (AKD) and CKD, with consensus definitions based on widely used clinical measures, the effective use of disease registries and electronic health records to identify and address areas of unmet need, the discovery of genetic variants that increase kidney disease

risk, the development of risk tools for individual risk prediction to guide clinical care, and the application of innovative technologies for the discovery of biomarkers of kidney disease.

CONSENSUS DEFINITIONS, CONCEPTUAL MODELS, AND CLASSIFICATION OF CHRONIC AND ACUTE KIDNEY DISEASE

The terms to describe AKD and CKD have evolved over the years. Consensus definitions were first provided in clinical practice guidelines in the early 2000s, a critical step to allow the clinical translation of epidemiologic research. The definition and classification for CKD was proposed by the

U.S. National Kidney Foundation Kidney Disease Outcome Quality Initiative in 2002,^{1,2} adopted by the international guideline group Kidney Disease Improving Global Outcomes (KDIGO) in 2005,³ and updated by KDIGO in 2012 and 2024.⁴ More detailed considerations for the definitions of kidney failure and CKD progression for use in clinical trials were provided by an international consensus group in 2020.^{4a}

The definition and classification of acute kidney injury (AKI) were proposed by the Acute Dialysis Quality Initiative in 2004,⁵ modified by the Acute Kidney Injury Network (AKIN) in 2007,⁶ and harmonized by KDIGO in 2011, which also provided a definition for AKD.^{7,8} New KDIGO guidelines for AKI and AKD are currently in development. The definitions and staging systems for CKD, AKD, and AKI are based on kidney measures that are frequently obtained in clinical practice and population studies (Table 18.1).

The conceptual models for the development, progression, and complications of AKD and CKD are similar.^{4,7} They include antecedents associated with an increased risk for the development of kidney disease, stages of disease, and complications of disease, including death (Fig. 18.1). Risks for the development of kidney disease may be categorized as susceptibility due to demographic and genetic factors or exposure to factors that might initiate kidney disease. Kidney disease is defined by abnormalities in structure or function, with abnormalities in kidney structure (damage) often, but not always, preceding abnormalities in function. Markers of kidney damage include pathologic findings on biopsy, abnormalities in urine sediment or imaging, proteinuria (including albuminuria), and alterations in tubular function. Glomerular filtration rate (GFR) is widely accepted as the best measure of overall kidney function. A GFR < 60 mL/min/1.73 m² is considered moderately decreased kidney function, and a GFR < 15 mL/min/1.73 m² is defined as kidney failure. Complications include conditions that occur more often in people with kidney disease compared with people without kidney disease. In addition to metabolic and hormonal disorders associated with CKD (e.g., anemia, mineral and bone disease, malnutrition, and neuropathy), complications include cardiovascular disease (CVD), drug toxicity, and a wide variety of other conditions, such as infections, cognitive impairment, and frailty. Death may result from kidney failure or complications.

Kidney disease is classified according to its duration, severity, cause, and prognosis.⁴ Duration of kidney disease longer than 3 months is defined as CKD, whereas duration less than or equal to 3 months is defined as AKD, of which AKI is the subset of cases that occur within 7 days. Duration can be established through repeat measurements or from corroboratory data, such as imaging or biopsy findings.⁹ Criteria for CKD include GFR < 60 mL/min/1.73 m² or markers of kidney damage. The severity of CKD is assessed by the level of GFR (stages G1–G5) and albuminuria (stages A1–A3), with end-stage kidney disease (ESKD) as the subset of stage G5 treated by dialysis and transplantation, collectively known as “kidney replacement therapy” (KRT). Causes of CKD are classified according to the presence or absence of systemic disease and location of pathologic and anatomic abnormalities within the kidney. Prognosis is

determined based on G- and A-staging and categorized as moderate, high, and very high risk (Fig. 18.2).

Criteria for AKI include a rise in the serum creatinine level or oliguria, and severity is staged according to peak value for the serum creatinine level or nadir value for urine output. Patients requiring acute KRT are classified as stage 3, the most severe stage. Pathophysiologic classification includes decreased kidney perfusion, obstruction of the urinary tract, and parenchymal kidney diseases (including acute tubular necrosis). Criteria for AKD include AKI, GFR < 60 mL/min/1.73 m², a decline in GFR by 35% or more or rise in serum creatinine by 50% or more from baseline, or markers of kidney damage, each present for < 3 months. Both AKI and AKD may occur in people with or without underlying CKD. Incomplete recovery of AKD can lead to new-onset CKD in those without underlying CKD or progression of CKD in those with underlying CKD. In addition, there is overlap among the risk factors, causes, and complications between AKD and CKD.

PRINCIPLES OF EPIDEMIOLOGY AND USE IN KIDNEY DISEASE RESEARCH

This chapter focuses on the prevalence, incidence, outcomes, and risk factors for kidney disease. Disease prevalence is the proportion of individuals in a given population with disease; disease incidence is the number of new cases that develop among individuals without prevalent disease over a given period. Disease prevalence thus depends on not only disease incidence but also survival and how long the disease persists. Outcomes refer to events of interest that may occur with greater frequency among affected individuals, and risk factors refer to characteristics that precede the onset of disease or outcome. It should be noted that much of the epidemiologic literature is observational, and thus observed associations among risk factors, kidney disease, and outcomes may be causal or noncausal. The Bradford Hill criteria provide guidelines for inferring causation and specify the minimal conditions that need to be met to support a causal relationship between the risk factor (exposure) and disease (outcome).¹⁰ Trends refer to changes in disease incidence, prevalence, or outcomes over time and may be due to changes in population demographics, risk factor distribution, risk factor control, or availability and effectiveness of treatment. The latter explanation is particularly relevant for trends in ESKD, which is defined by treatment with KRT.

RECENT CONTRIBUTIONS OF EPIDEMIOLOGY TO NEPHROLOGY

Epidemiologic studies underlie many of the important advances in kidney disease in the past 2 decades. The development of GFR-estimating equations and recognition of albuminuria as a risk factor for adverse outcomes were central to the definition and staging of CKD, which in turn enabled studies of the prevalence of CKD, identification of CKD as a risk factor for CVD, and recognition of CKD as a global public health problem.^{11,12} In addition, estimated GFR (eGFR) and albuminuria are now widely used as prognostic markers in clinical practice and as surrogate markers of kidney disease

Table 18.1 Definitions and Classification of Chronic Kidney Disease (CKD), Acute Kidney Disease (AKD), and Acute Kidney Injury (AKI)

Parameter	CKD	AKD	AKI
Definition			
Functional criteria	GFR < 60 mL/min/1.73 m ²	AKI, or GFR <60 mL/min/1.73 m ² , or Decrease in GFR by ≥35% times baseline, or Increase in SCr by >50% times baseline	Increase in serum creatinine level (SCr) by 50% within 7 days or Increase in SCr by 0.3 mg/dL within 2 days, or Oliguria (urine output <0.5 mL/kg/h for >6 hours)
Structural criteria	Kidney damage	Kidney damage	No criteria
Duration	>3 months	≤3 months	≤1 week
Classification			
Acute pathophysiology	NA	Same as AKI	Decreased kidney perfusion, obstruction of the urinary tract, parenchymal kidney disease other than acute tubular necrosis, and acute tubular necrosis
Cause	Presence or absence of systemic disease and location in the kidney of pathologic-anatomic abnormalities (e.g., glomerular, tubulointerstitial, vascular, and cystic)	Same as CKD	Not specified separately
Severity (stage)	Stages based on GFR (G) and albuminuria (A) categories, and related terms ^a : G1 ≥90: Normal or high G2 60–89: Mildly decreased ^b G3a 45–59: Mildly to moderately decreased G3b 30–44: Moderately to severely decreased G4 15–29: Severely decreased G5 <15 or KRT (kidney failure) A1 <0: Normal to mildly increased A2 30–300: Moderately increased ^b A3 >300: Severely increased ^b	Not specified separately	Stages based on SCr or urine output: 1: SCr ≥1.5–1.9 times baseline or >0.3- mg/dL increase or urine output <0.5 mL/kg/h for 6–12 hours 2: SCr ≥2.0–2.9 times baseline or urine output <0.5 mL/kg/h for ≥12 hours 3: SCr ≥3.0 times baseline or to ≥4.0 mg/ dL or KRT In patients aged <18 years, decrease in eGFR to <35 mL/min/1.73 m ² or urine output <0.3 mL/kg/h for ≥24 hours or anuria for ≥12 hours
Risk categories	Moderate (G1–G2, A2; G3a, A1) High (G1–G2, A3; G3a, A2; G3b, A1) Very high (G3a, A3; G3b, A2–A3; G4, A1–A3; G5, A1–A3)	Not specified separately	Stages 1, 2, and 3
Treatment	KRT included in stage 5, defined as ESKD	Not specified separately	KRT included in stage 3

^aUnits for GFR, mL/min/1.73 m²; units for albuminuria categories, AER mg/day, ACR mg/g. In the absence of evidence of kidney damage, GFR category G1 or G2 does not fulfill the criteria for CKD. In the absence of GFR <60, albuminuria category A1 does not fulfill the criteria for CKD.

^bTerms for categories G2 and A2 are relative to young adult levels. Category A3 includes nephrotic syndrome (albumin excretion usually >2200 mg/day [ACR >2220 mg/g; >220 mg/mmol]).

ACR, Albumin-to-creatinine ratio; AER, albumin excretion rate; CKD, chronic kidney disease; ESKD, end-stage kidney disease; GFR, glomerular filtration rate; KRT, kidney replacement therapy; NA, not applicable; SCr, serum creatinine concentration.

Adapted from Kidney Disease: Improving the Improving Global Outcomes (KDIGO) Acute Kidney Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney Int Suppl.* 2012;2:1–138; and Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3:1–150.

outcomes in clinical trials of treatment for kidney disease.¹³ Advances in genetic epidemiology have revolutionized our understanding of both common and uncommon diseases. Common variations among individuals of African ancestry in a single gene encoding apolipoprotein L1 (APOL1) are

now recognized to increase the risk for ESKD.^{14–17} Similarly, large cohort studies have demonstrated that the presence of sickle cell trait increases the risk for both CKD and ESKD, increasing our understanding of racial disparities in kidney disease.^{18,19} Genome-wide association studies in

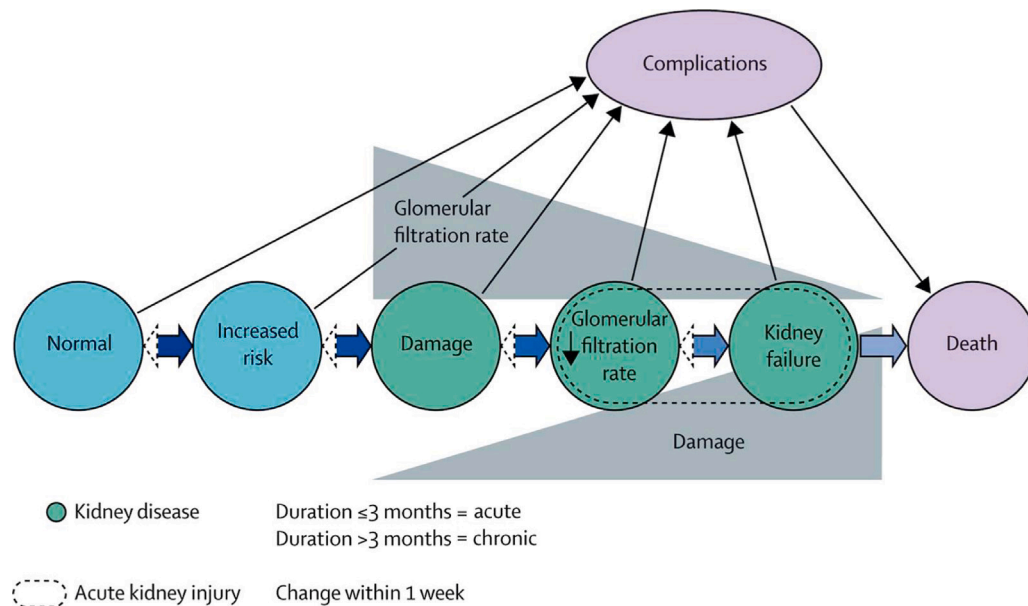


Fig. 18.1 Factors associated with increased risk of kidney disease (blue), stages of disease (green), and complications (including death; purple). (From Eckardt KU, Coresh J, Devuyst O, et al. Evolving importance of kidney disease: from subspecialty to global health burden. *Lancet*. 2013;382:158–169.)

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30-300 mg/g 3-30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (mL/min/1.73 m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60-89			
	G3a	Mildly to moderately decreased	45-59			
	G3b	Moderately to severely decreased	30-44			
	G4	Severely decreased	15-29			
	G5	Kidney failure	<15			

Fig. 18.2 Chronic kidney disease staging. Classification is based on the identification of cause (C), glomerular filtration rate (GFR) category (G), and albuminuria (A). This is collectively known as “CGA staging.” (From Kidney Disease: Improving Global Outcomes [KDIGO] CKD Work Group. Clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.)

membranous and immunoglobulin A (IgA) nephropathy have provided strong evidence of heritability and a role for the immune system in the development of disease.²⁰⁻²⁴ Rare genetic variants have been identified as the cause of several monogenic diseases, such as polycystic kidney disease, familial nephrotic syndrome, and congenital anomalies of the kidney and urinary tract.^{25,26} Global collaboration, coupled with the increasing availability of large databases with rigorously defined clinical characteristics, frequent ascertainment of kidney measures, and genomic, proteomic, and metabolomic data will allow continued advances in research, clinical practice, and public health.

CHRONIC KIDNEY DISEASE

METHODOLOGIC CONSIDERATIONS

Obtaining accurate and comparable estimates of CKD burden across world regions is difficult for several reasons. Identification of disease requires testing for kidney disease measures (e.g., serum creatinine or cystatin C for eGFR, urine albumin, and urine total protein), which are not uniformly performed. Differences across populations may reflect differences in laboratory testing practice because CKD in its early stages is typically asymptomatic, with low awareness of disease.^{27,28} For

example, if only eGFR is ascertained, prevalence estimates reflect CKD stages G3–G5 and will be considerably lower than if albuminuria is also ascertained, enabling prevalence estimates of CKD stages G1–G5 and A1–A3. Two populations with equal proportions of CKD also may have different prevalence estimates if one performs near-universal laboratory testing and the other screens only high-risk individuals. In addition, because CKD is defined using kidney disease measures, assays must be standardized to derive comparable estimates. Standardization of creatinine assays has been achieved,²⁹ and substantial progress has been made for cystatin C³⁰; however, standardization of urine albumin assays is still ongoing.³¹ Finally, estimating equations for GFR are not consistently applied, and there is known variation in CKD prevalence when, for example, the Modification of Diet in Renal Disease or different Chronic Kidney Disease Epidemiology Collaboration equations are used.^{32,33} Progress in CKD awareness, testing, definitions, and measurement should enable substantial progress in the field in the coming decade.

PREVALENCE

Estimates of the prevalence of CKD worldwide vary but generally range between 8% and 16% in different populations.³⁴ The most accurate CKD prevalence estimates come from nationally representative surveys, such as the National Health and Nutritional Examination Survey (NHANES) in the United States. In this sampling design, people are selected from the general population and undergo detailed physical and laboratory examinations. Estimates of disease prevalence are obtained by weighting sample proportions by inverse probability of selection from the general population. In NHANES 2017–2020, the estimate of U.S. prevalence of CKD based on a one-time measurement of eGFR creatinine (Cr) and urine albumin-to-creatinine ratio was 14.0%, representing 8.4% for stages G1 and G2 with albuminuria A2 or A3 and 5.6% for stage G3 or G4.³⁵ It should be noted that this study did not assess the chronicity of abnormalities of these measures and thus may have overestimated their prevalence. However, other considerations suggested an underestimation of prevalence of CKD. Markers of kidney damage other than albuminuria were not assessed. People with CKD, stage G5—those undergoing KRT and those with an eGFR < 15 mL/min/1.73 m²—were not included in the NHANES estimates. In addition, NHANES does not include people living in institutionalized settings or who are involved in the military.

Trends in the prevalence of CKD suggest relative stability over time (13.3% in 2005–2008 to 14.1% in 2017–2020), which is encouraging when one considers the significant increases in the prevalence of diabetes mellitus and obesity.^{35,36,38}

In Europe, no multinational survey data exist, so studies typically combine data to estimate age- and sex-adjusted CKD prevalence.^{39,40} A recent study of 2.4 million patients with CKD obtained from digital health care systems in nine countries (including Canada and Israel) reported a pooled prevalence of 10%, with heterogeneity in albuminuria testing (a key component of CKD identification) ranging from 11% to >90%.⁴⁰ In a study combining 19 general population cohorts from 13 European countries, there was large heterogeneity by country, even when stratified by diabetes, hypertension, and obesity status, with lower rates of CKD stages G3–G5 in central Italy (1%) but higher in northeast Germany (5.9%).³⁹ Similarly, the prevalence of CKD stages G1–G5 ranged from 3.3% in Norway to 17.3% in northeast Germany. **Fig. 18.3**

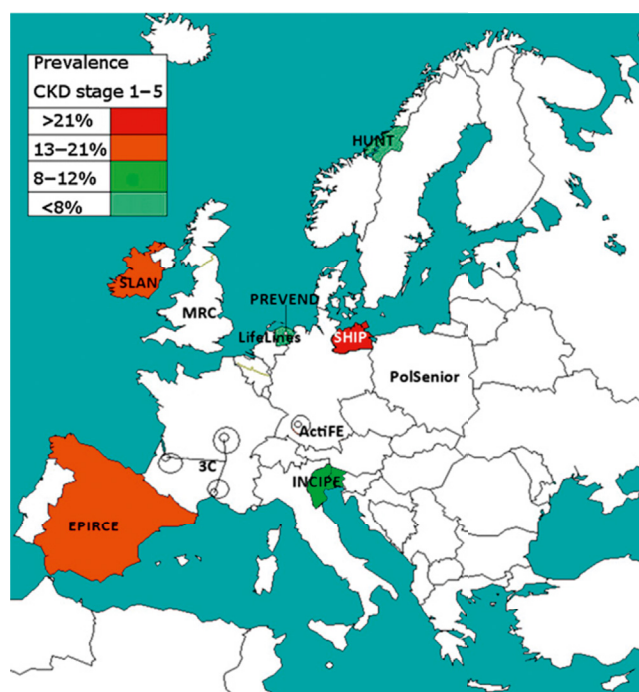


Fig. 18.3 Adjusted prevalence of chronic kidney disease (CKD) stages 1–5 in European populations aged 45–74 years, in isotope dilution mass spectrometry (IDMS) studies. Prevalence was age and sex adjusted to the EU27 population of 2005. The study names in the uncolored regions are studies that used non-IDMS-standardized creatinine or studies that recruited subjects aged 50+ years. (From Brück K, VS, Gambaro G, et al. CKD prevalence varies across the European general population. *J Am Soc Nephrol.* 2016;27:2135–2147.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

indicates the differences by region in the population aged 45 to 74 years and highlights the relatively localized source populations, which may not be representative of all of Europe. Other limitations noted by the authors included the heterogeneity in laboratory methods, with not all creatinine assays being isotope dilution mass spectrometry standardized, and differences in dietary habits, with certain high-protein diets typical of northern Germany known to influence serum creatinine (Cr), resulting in an underestimate of eGFRcr.

Global estimates of CKD are also heterogeneous (**Table 18.2**; see also **Chapter 73**).^{4,12,41,42} A systematic analysis of 33 studies from 32 countries (nearly 50% of the world's population) has found the crude prevalence of CKD stages G1–G5 ranges in men from 4.5% in South Korea to 25.7% in El Salvador, and in women from 4.1% in Saudi Arabia to 16.0% in Singapore.⁴¹ Underlying studies differed in the creatinine measurement method, eGFRcr equation used, and protocol for assessing albuminuria. Noting these caveats, the authors estimated the age-standardized global prevalence of CKD in 2010 as 10.4% in men and 11.8% in women. They also reported substantial differences by gross national product, with high-income countries estimated to have lower CKD prevalence (8.6% and 9.6% in men and women, respectively) than low- and middle-income countries (10.6% and 12.5%). Of note, a more recent study of the prevalence of CKD in Asian countries offered different estimates for 2020, ranging from 7.0% in South Korea to 34.3% in Singapore.⁴³

Diabetes and hypertension are considered the major causes of CKD globally.³⁴ In the U.S. population with diabetes

Table 18.2 Global Estimates of Chronic Kidney Disease^a

		High-Income Countries				Low- and Middle-Income Countries			
		CKD Stages 1-5		CKD Stages 3-5		CKD Stages 1-5		CKD Stages 3-5	
Age (Years)	Men	Women	Men	Women	Men	Women	Men	Women	Men
Prevalence, % (95% CI)									
20-29	3.7 (2.7-5.1)	5.3 (3.8-6.3)	0.7 (0.3-1.4)	0.9 (0.4-1.6)	7.3 (6.4-9.0)	6.6 (6.2-7.3)	3.0 (1.7-5.4)	2.0 (1.4-3.3)	
30-39	5.0 (4.0-6.0)	5.9 (4.4-6.9)	1.3 (0.7-2.1)	1.6 (0.9-2.6)	8.1 (6.8-10.3)	9.0 (8.6-9.7)	3.1 (1.9-5.0)	3.1 (2.1-5.1)	
40-49	6.8 (5.5-8.2)	7.7 (5.9-9.0)	2.1 (1.4-3.1)	3.2 (2.0-4.8)	10.2 (9.0-12.6)	11.5 (11.0-12.7)	3.0 (1.8-6.2)	4.0 (2.5-7.7)	
50-59	10.2 (8.6-12.2)	11.1 (8.6-13.8)	4.6 (3.2-6.7)	7.4 (5.2-10.2)	12.0 (10.4-15.1)	15.7 (14.7-17.8)	4.9 (3.5-8.1)	6.7 (4.5-11.7)	
60-69	16.0 (13.6-18.1)	15.6 (12.6-18.9)	9.4 (7.8-11.6)	12.2 (9.6-15.8)	16.3 (14.7-20.0)	21.3 (19.6-24.9)	9.7 (6.1-15.6)	13.1 (9.5-19.7)	
≥70	28.1 (23.4-33.0)	28.9 (22.7-34.3)	22.7 (20.0-26.4)	28.5 (26.1-31.5)	20.6 (19.4-24.1)	28.4 (26.3-32.7)	11.8 (9.0-17.6)	17.3 (12.6-27.2)	
Total	10.1 (8.8-11.1)	12.1 (9.9-13.7)	5.4 (4.6-6.5)	8.6 (6.9-10.7)	10.2 (9.1-12.4)	12.1 (11.6-13.3)	4.3 (2.9-7.1)	5.3 (3.8-8.2)	
Age-standardized	8.6 (7.3-9.8)	9.6 (7.7-11.1)	4.3 (3.5-5.2)	5.7 (4.4-7.6)	10.6 (9.4-13.1)	12.5 (11.8-14.0)	4.6 (3.1-7.7)	5.6 (3.9-9.2)	
Absolute Numbers (in Thousands; 95% CI)									
20-29	3453 (2485-4718)	4647 (3307-5500)	694 (288-1275)	800 (307-1570)	37,121 (32,209-45,651)	32,054 (30,213-35,932)	14,998 (8471-27,644)	9863 (6746-16,328)	
30-39	4699 (3802-5684)	5298 (4000-6235)	1221 (659-1957)	1440 (712-2514)	33,274 (27,740-42,285)	35,920 (34,256-38,727)	12,499 (7803-20,591)	12,265 (8185-20,402)	
40-49	6326 (5158-7691)	7120 (5519-8386)	2004 (1290-2934)	2990 (1731-4700)	35,322 (31,190-43,673)	39,012 (37,165-43,201)	10,253 (6328-21,335)	13,495 (8579-26,111)	
50-59	8531 (7149-10,157)	9752 (7541-12,136)	3874 (2625-5549)	6485 (4218-9418)	29,618 (25,640-37,287)	38,337 (35,959-43,651)	12,167 (8612-19,988)	16,355 (10,900-28,611)	
60-69	9579 (8155-10,849)	10,425 (8411-12,690)	5608 (4694-6963)	8291 (6329-11,264)	22,434 (20,338-27,637)	31,136 (28,731-36,396)	13,420 (8430-21,497)	19,232 (13,883-28,848)	
≥ 70	15,699 (13,066-18,416)	24,426 (19,150-29,003)	12,690 (11,136-14,745)	24,065 (21,769-27,161)	19,606 (18,459-22,925)	33,637 (31,150-38,706)	11,176 (8541-16,673)	20,509 (14,912-32,240)	
Total	48,285 (42,349-53,284)	61,669 (50,394-69,929)	26,091 (22,014-31,078)	44,071 (35,296-54,831)	177,375 (159,157-215,924)	210,096 (200,787-231,704)	74,513 (50,324-123,460)	91,719 (66,697-143,121)	

^aAge-specific and age-standardized prevalence estimates and absolute numbers of men and women with CKD in high-, low-, and middle-income countries.

CI, Confidence interval; CKD, chronic kidney disease.

From Mills KT, Xu Y, Zhang W, et al. A systematic analysis of worldwide population-based data on the global burden of chronic kidney disease in 2010. *Kidney Int.* 2015;88(5):950-957.

mellitus, an estimated 38% had CKD in 2017–2020; in the population with hypertension, an estimated 21.5% had CKD, and in the population with obesity, an estimated 16.5% had CKD.³⁵ The prevalence of CKD by specific cause is difficult to estimate, but if the prevalence of diabetes mellitus in the United States is approximately 10%⁴¹ and more than one-third of people with diabetes mellitus have CKD, then the prevalence of diabetic kidney disease in the U.S. population may be close to 3% to 4%. As trends in the prevalence of kidney disease in those with diabetes mellitus have been estimated to be fairly flat,⁴⁴ trends in the prevalence of diabetic kidney disease in the general population likely follow the increasing trends in the prevalence of diabetes.⁴⁵

INCIDENCE

Compared with estimates of CKD prevalence, information on the incidence of CKD is relatively sparse. Calculation of the incidence—the development of new cases of CKD—requires a population-representative sample, longitudinal follow-up, and regular and comparable ascertainment of kidney measures in everyone. The incidence of disease stemming from clinical populations may not be valid because higher-risk individuals are more likely to receive testing. Noting this limitation, equations for the 5-year risk of incident eGFR <60 mL/min/1.73 m² have been developed.⁴⁶ Covariates include age, sex, race/ethnicity, eGFR, history of CVD, ever smoker, hypertension, body mass index, and albuminuria concentration; in persons with diabetes, additional covariates include diabetes medications, hemoglobin A1c, and their interaction. The risk equations had a median C statistic of 0.845 (25%, 75% range, 0.789–0.890) in the cohorts without diabetes and 0.801 (25%, 75% range, 0.750–0.819) in the cohorts with diabetes. On the other hand, the incidence of disease stemming from population-based cohorts may underestimate CKD incidence due to loss of follow-up and relatively infrequent study visits (e.g., visits every 3 or 5 years).

In the United States, a few long-running prospective cohorts have provided estimates of incident CKD, but the frequency of assessment of kidney measures differed. The Atherosclerosis Risk in Communities study, a cohort of middle-aged Black and White individuals aged 45 to 65 years of age, from 4 U.S. communities, estimated an incidence of 10.4 cases of CKD stages G3–G5/1000 years of follow-up, or a 7.3% incidence over 8.8 years when eGFRcr was assessed every 3 years.⁴⁷ The Framingham study, a community-based study of predominantly White individuals with a relatively low rate of diabetes (2.7% vs. 11.4% in the Atherosclerosis Risk in Communities study) and a mean age of 43 years, reported 9.4% of 2585 individuals developed CKD stages G3–G5 in a subsequent visit 18.5 years later. In the Cardiovascular Health Study, a community-based longitudinal study of older adults with a mean age of 72 years, 10% developed CKD stages G3–G5 by year 7 when assessed using eGFRcr, and 19% developed CKD stages G3–G5 when assessed using eGFRcys. In each of these studies, the rate of incident CKD was higher with older age. Notably, none assessed the incidence of albuminuria, so they likely underestimated the incidence of CKD.

Racial differences in incident CKD also exist. In the Coronary Artery Risk Development in Young Adults cohort, a study of Black and White individuals aged 18–30 years, the risk of developing CKD stages G3–G5 (in this study, assessed from eGFRcr and requiring an accompanied 25%

decline in eGFR) at 10, 15, or 20 years after baseline was 2.6 times higher in Black participants compared with White participants.⁴⁸ Similarly, there were large racial disparities in the Multi-Ethnic Study of Arteriosclerosis (mean age, 60 years), in which Black participants had a more than threefold higher risk of developing CKD stages G3 to G5 (in this study, assessed from eGFRcys and required a ≥ 1 mL/min/1.73 m² decline in eGFR) compared with White participants in a population with baseline eGFR > 90 mL/min/1.73 m² (3.4 cases/1000 years vs. 0.9 cases/1000 years). Incidence rates were higher among participants with eGFR between 60 and 90 mL/min/1.73 m², and racial differences persisted, albeit slightly attenuated (26.3 cases/1000 years vs. 18.5 cases/1000 years in Black and White participants, respectively). When modeled as incidence over a lifetime using NHANES and registry data, racial disparities again were present, with African Americans projected to develop CKD stages G3–G5 at younger ages and greater disparities in the incidence of more severe disease (Fig. 18.4).⁴⁹

In Europe, the incidence of CKD assessed using eGFRcr and urine albumin excretion >30 mg/day has been reported in the PREVEND cohort, a population-based cohort that oversampled for people with urine albumin concentrations ≥ 10 mg/L on a first-morning void urine sample.⁵⁰ Participants were followed over 10 years at 5 separate visits. The incidence of developing CKD stages G1–G5 was 18.6/1000 person-years; the incidence of developing albuminuria was 13.4/1000 person-years, and the incidence of developing decreased GFR was 5.8/1000 person-years. In an administrative cohort, the incidence of serum creatinine ≥ 1.7 mg/dL (eGFR ranging from 37–57 mL/min/1.73 m² for men and 28–43 mL/min/1.73 m² for women) was 1701 cases/million population in Southampton, United Kingdom, with much higher rates of CKD among older compared with younger inhabitants.⁵¹

OUTCOMES

CKD has been increasingly recognized as a top public health priority due to its strong association with adverse outcomes. In some areas of the world, CKD is one of the top five causes of death. The Global Burden of disease estimated that 1.2 million people died from CKD in 2017.² Lower GFR and higher albuminuria levels result in a significantly diminished life expectancy at all ages (Fig. 18.5).²⁸ Part of this increase in mortality is exerted through the higher risk of kidney failure, both untreated and treated by dialysis or transplantation.⁵² In addition, earlier stages of CKD are associated with increased risk of death, AKI, CKD progression, CVD, heart failure, and even some types of cancer.^{53–57} In an earlier analysis of general population cohorts and in an updated meta-analysis of 27.5 Million individuals from 114 global cohorts the heat map pattern of risk is consistent across 10 common outcomes (Fig. 18.6).⁵³ For AKI and other adverse outcomes, risk increases with lower GFR and higher albuminuria are generally independent, with each parameter carrying risk, and higher risk with higher stages of disease.^{58,59} Risk relationships have been observed in low-risk and high-risk groups for ESKD and mortality.^{60–63}

There is intense interest in defining earlier outcomes of CKD progression, with the goal of enhancing drug development and clinical prevention.⁶⁴ Provided that there are no acute effects on eGFR, decline in eGFR by 30% to 40% is now accepted by the U.S. Food and Drug Administration as

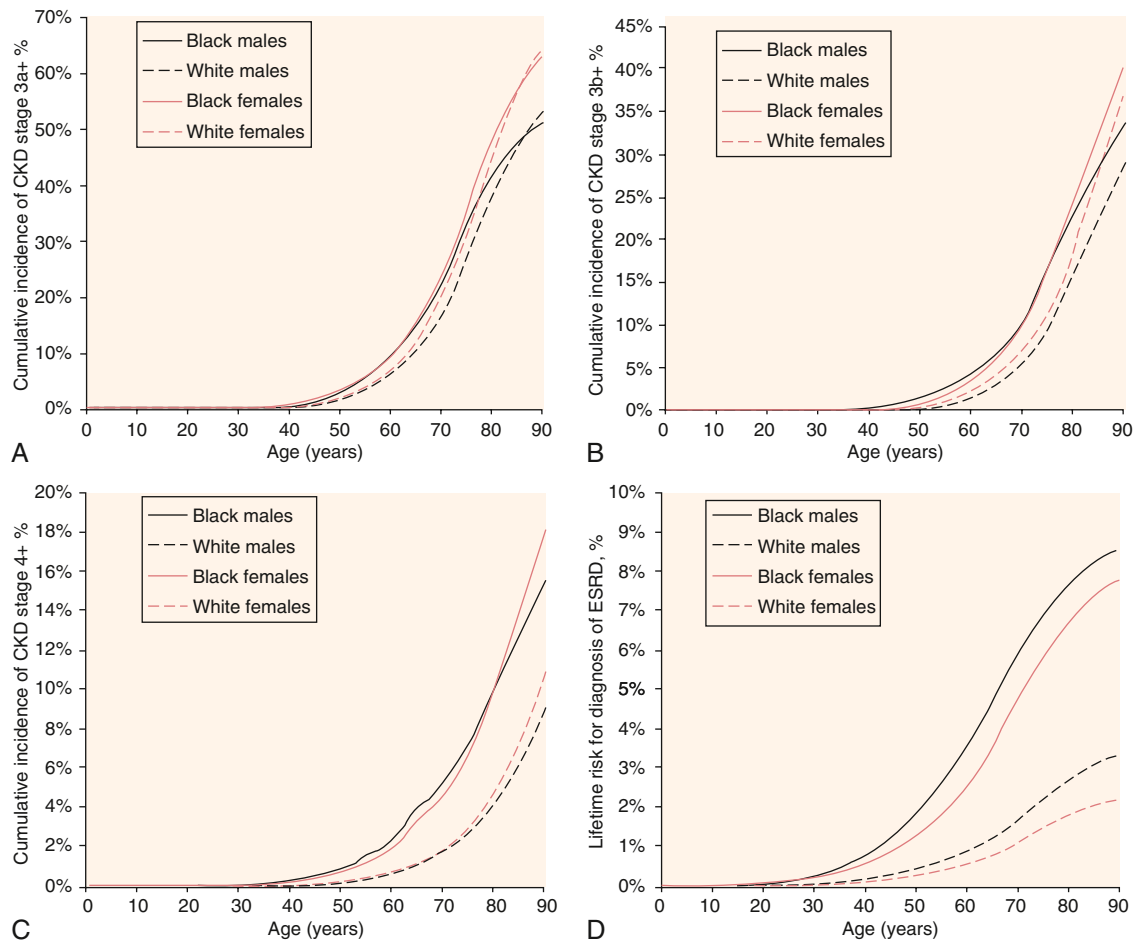


Fig. 18.4 Cumulative incidence from birth by race and sex of chronic kidney disease (CKD) stage 3a (A), 3b (B), 4 (C), and end-stage renal disease (ESRD; D). (From Grams ME, Chow EK, Segev DL, et al. Lifetime incidence of CKD stages 3–5 in the United States. *Am J Kidney Dis.* 2013;62:245–252.)

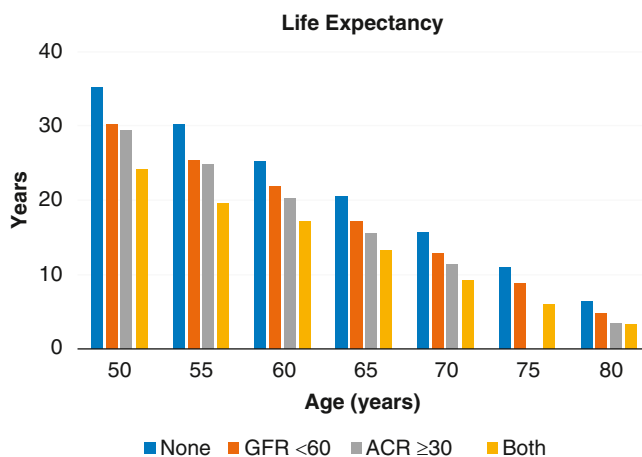


Fig. 18.5 Life expectancy of NHANES participants with or without abnormalities in chronic kidney disease (CKD) measures, 1999–2011. GFR <60 mL/min/1.73 m² or ACR ≥30 mg/g indicate CKD. ACR, Urine albumin-to-creatinine ratio; GFR, estimated glomerular filtration rate; NHANES, National Health and Nutrition Examination Survey. (Adapted from U.S. Renal Data System. 2016 USRDS annual data report: epidemiology of kidney disease in the United States. <[https://www.ajkd.org/article/S0272-6386\(16\)30703-X/fulltext](https://www.ajkd.org/article/S0272-6386(16)30703-X/fulltext)>; 2016. Accessed 02.09.2024.)

a surrogate outcome for CKD progression, previously assessed by doubling of the serum creatinine level (equivalent to a 57% decline in eGFR) or the development of ESKD.¹³ Slope of GFR and changes in albuminuria are also associated with ESKD and death,^{65–68} and these endpoints, as well as remission of the nephrotic syndrome, are used as a surrogate endpoint in trials of some glomerular diseases.^{69,70}

Kidney measures are not the only risk factors for adverse outcomes, and there is a growing literature on the use of other factors to explain variations in the risk of progression to ESKD within stages of CKD.^{71–73} For a given level of GFR or albuminuria, older adults are less likely to experience ESKD than their younger counterparts, possibly due to treatment preference (e.g., refusal of dialysis or transplantation as a therapy) or because of the competing event of death (i.e., older adults with a given level of GFR or albuminuria are relatively more likely to die than progress to ESKD).⁷⁴ Moreover, the absolute risks of ESKD and death may vary based on indications for selection into a study population; research study populations often report a higher risk of ESKD than death, whereas clinical population information drawn from medical records data often shows a higher risk of death than ESKD.^{75,76} Regional variation in the absolute risk of progression to ESKD has also been reported. In a validation of a kidney failure risk equation calculator in

			CKD A-stages by urine albumin-creatinine ratio (ACR) ranges, mg/g				
			A1		A2		A3
			<10	10-29	30-299	300-999	1000+
CKD G-stage by eGFRcr ranges, ml/min/1.73 m ²	G1	90-104	REFERENCE ACM Hosp PAD MI Stroke CVM HF AF AKI KFRT	1.3 1.1 1.3 1.3 1.3 1.3 1.3 1.2 1.4 1.8	1.8 1.3 1.8 1.6 1.9 1.9 2 1.5 2.1 4.3	2.6 1.5 2.2 2.4 2.8 2.7 2.8 1.9 3.2 12	3.1 1.7 3.2 3.1 4.3 3.6 4.2 2.3 5 43
			1 1 1.1 1.1 1 1 1.1 1 1.6 2.3	1.3 1.1 1.3 1.3 1.3 1.4 1.4 1.2 2.2 4.9	1.7 1.3 1.6 1.7 1.8 1.7 1.9 1.4 3.1 10	2.2 1.5 2.2 2.2 2.5 2.4 2.7 1.7 4.3 27	2.8 1.8 3.1 3 3.8 3.2 4.2 2.2 6.7 85
			1.3 1.3 1.4 1.4 1.5 1.4 1.6 1.2 3.5 13	1.6 1.3 1.7 1.6 1.7 1.7 1.8 1.3 4 19	2 1.5 2 1.9 2.1 2.2 2.4 1.5 5.1 37	2.4 1.7 2.8 2.3 2.9 2.8 3.4 1.8 6.9 89	3.1 2.1 3.7 2.9 4.2 3.8 5 2.4 9 236
			1.8 1.5 1.9 1.6 2 2 2.2 1.4 5.6 50	2 1.5 2 1.7 1.9 2.3 2.5 1.5 5.9 58	2.5 1.6 2.4 2 2.5 2.8 3.1 1.7 6.8 115	3.2 1.9 3.2 2.4 3.6 3.7 4.2 2 8.6 240	3.9 2.3 4.3 3 6 4.6 5.5 2.4 11 463
			2.8 1.8 2.7 1.8 3.3 3.2 3.6 1.9 8.3 283	2.8 1.8 3.1 2.1 3.3 3.1 3.5 1.8 8 301	3.3 1.9 3.1 2.1 3.8 3.6 4.1 2 8.5 443	4.1 2.4 4.2 2.7 5.7 5 5.8 2.6 9.9 796	5.6 2.8 5.1 3 8.1 6.6 8.1 3 10 1253
			4.6 2.7 4.6 3.2 9.1 6.1 5.1 2.6 8.5 770	6 2.8 5.6 2.8 9 6.4 5.7 2.5 11 1040	6.3 3 4.8 2.9 9.6 6.4 5.8 3.1 7.9 1618	6 3.2 6 3.2 13 7.3 7.9 3.6 5.5 2297	7 3.8 6 3.8 14 8.2 9.9 4.2 5.7 2547

Fig. 18.6 Adjusted hazard ratios of 10 outcomes as a function of CKD eGFR (G-) and albuminuria (A-) stages by heat map colors of risk stages (green = low risk, yellow = moderately increased risk, orange = high risk, and red = very high risk). Outcome abbreviations (number of events) : all-cause mortality (ACM, n=2.6 M), Hospitalizations (Hosp, n=8.4 M), myocardial infarction (MI, n=0.451 M), Stroke (n=0.461 M), peripheral arterial disease (PAD, n=0.378 M), cardiovascular mortality (CVM, n=0.776 M), heart failure (HF, n=1.132 M), atrial fibrillation (AF, n=1.069 M), acute kidney injury (AKI, n=1.408), kidney failure replacement therapy (KFRT, n=0.159 M). Numbers reflect the adjusted hazard ratio compared with the reference cell in a separate analysis for each of the ten outcomes. Adjustment variables included: age, sex, smoking status (current, former, never), systolic blood pressure, total cholesterol, high-density lipoprotein cholesterol, body mass index, use of antihypertensive medications, and a history of diabetes, coronary heart disease, stroke, heart failure, atrial fibrillation, peripheral artery disease, cancer, and chronic obstructive pulmonary disease when relevant. The percentile shaded the darkest green color corresponds to the proportion of cells in the grid without chronic kidney disease, and the percentile shaded the darkest red color corresponds to the proportion expected to be at the highest risk for adverse outcomes. In this manner, the numbers of green and red cells are consistent across outcomes, but the patterns are allowed to differ. We omitted the eGFRcr category of 105+ mL/min/1.73 m² from this figure since this category displays an increased risk for some outcomes. (Data from Writing Group for the CKD Prognosis Consortium. Grams ME, Coresh J, Matsushita K, et al. Estimated glomerular filtration rate, albuminuria, and adverse outcomes: An individual-participant data meta-analysis. *JAMA*. 2023;3:330(13):1266-1277.)

31 international cohorts, only the region of cohort was observed to improve calibration for a model that included age, sex, eGFR, and albuminuria.⁷¹ Therefore a regional calibration factor was introduced that incorporated a lower baseline risk by 32.9% at 2 years and 16.5% at 5 years for countries outside North America (see kidneyfailure.risk.com) (Fig. 18.7).⁷² Some of this risk differential may be due to the resources allotted to providing treatment rather than underlying characteristics of the countries themselves.⁷⁷

A lower GFR has also been associated with concomitant bone, metabolic, endocrine, and hematologic abnormalities.⁷⁸ Well-known pathophysiologic mechanisms underlie the associations of hyperphosphatemia, hyperparathyroidism, metabolic acidosis, and anemia with a lower GFR.⁷⁹ The relationship between these abnormalities and albuminuria is less consistent, but the presence of albuminuria does appear to impart a higher risk of hyperparathyroidism and anemia over and above the eGFR.

RISK FACTORS

Age is one of the strongest risk factors for the development of CKD. The prevalence of CKD stages G1 to G5 has been reported as nearly 40% in those older than 70 years and >50% in those 75 years and older in the United Kingdom.⁸⁰ Although some have dismissed age as a risk factor, suggesting that nephron loss is a normal part of aging,^{81,82} associations between a lower eGFR and higher albuminuria and adverse outcomes, including both death and ESKD, persisted in all subgroups of age in an individual-level meta-analysis of 2 million participants from 46 cohort studies.⁶³ On the other hand, among people with stages G3 to G5 CKD, older age is associated with a lower risk for ESKD, which may be due to the higher risk of mortality, both overall and attributable to CKD at an older age.^{71,72} The heat-map pattern of risk persisted in an updated meta-analysis of 5 Million people.^{71,72}

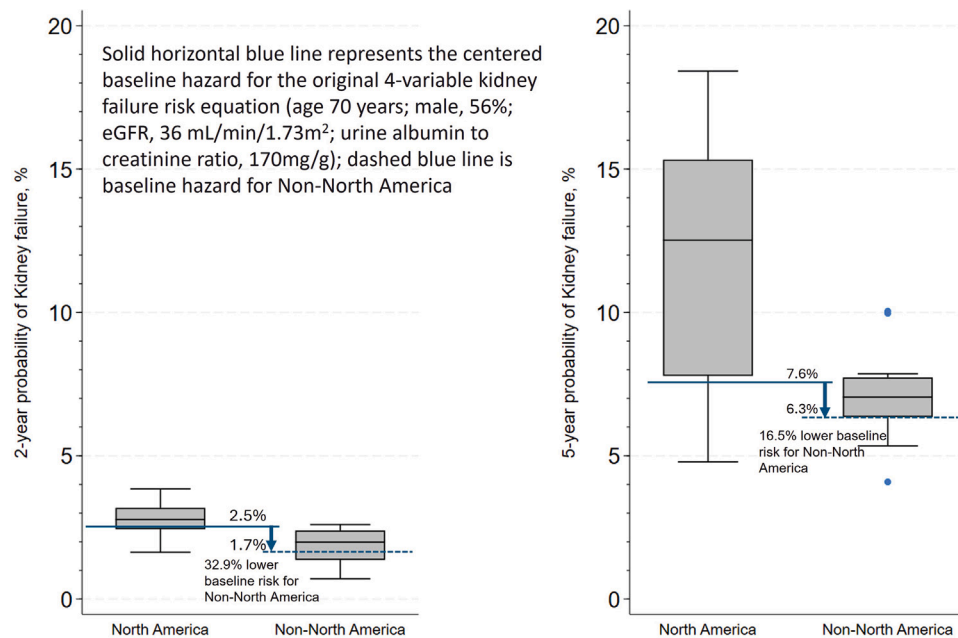


Fig. 18.7 Refit baseline hazard of original kidney failure risk equation (KFRE) at 2 and 5 years stratified by region. Box plots show the the centered baseline hazard for the original four-variable kidney failure risk equation (age, 70 years; male, 56%; eGFR, 36 mL/min/1.73 m²; urine albumin-to-creatinine ratio (ACR), 170 mg/g) in cohorts within and outside North America. The 25 cohorts included were: AASK, African American Study of Kidney Disease and Hypertension; ARIC, Atherosclerosis Risk in Communities; BC CKD, British Columbia Chronic Kidney Disease; CCF, Cleveland Clinic Foundation; CRIB, Chronic Renal Impairment in Birmingham; CRIC, Chronic Renal Insufficiency Cohort; eGFR, estimated glomerular filtration rate; GCKD, German CKD; GLOMMS, Grampian Laboratory Outcomes, Morbidity and Mortality Studies; HUNT, Nord Trøndelag Health Study; ICES-KDT, Institute for Clinical Evaluative Sciences, Provincial Kidney, Dialysis, and Transplantation; MASTER-PLAN, Multifactorial Approach and Superior Treatment Efficacy in Renal Patients With the Aid of a Nurse Practitioner; MDRD, Modification of Diet in Renal Disease; MMKD, mild to moderate kidney disease; NZDCS, New Zealand Diabetes Cohort Study; REGARDS, Reasons for Geographic and Racial Differences in Stroke Study; RENAAL, Reduction of Endpoints in Non-insulin Dependent Diabetes Mellitus With the Angiotensin II Antagonist Losartan; SRR-CKD, Swedish Renal Registry CKD; VA CKD, Veterans Administration CKD. (Based on data from Tangri N, Grams ME, Levey AS, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA*. 2016;315:164–174.)

Sex is also associated with CKD, although the relations are a bit nuanced. Women are at a slightly higher risk for the development of CKD stage G3; however, later stages of CKD are more common in men, suggesting that men may be at higher risk for CKD progression.^{49,83} The relation between eGFR and albuminuria and adverse outcomes was examined by sex in an individual-level meta-analysis of 2 million participants. Men had a higher absolute risk of ESKD, death, and cardiovascular mortality at all levels of eGFR and albuminuria, but the relation between eGFR and albuminuria and adverse outcomes was similar between the sexes.⁶⁰

Compared with Americans of primarily European ancestry, African Americans have a much higher risk of later-stage CKD and CKD progression. These racial differences are only partially explained by differences in socioeconomic or clinical risk factors.^{84–86} In a nationally representative cohort of adults older than 45 years, the prevalence of an eGFRcr of <60 mL/min/1.73 m² was 8.6% among Black individuals and 5.3% among non-Black individuals, and prevalence of eGFRcr < 30 mL/min/1.73 m² was 1.5% among Black individuals and 0.4% among non-Black individuals when using the CKD-EPI 2021 race-free equations.⁸⁸ The use of different estimating equations or different filtration markers results in slightly different prevalence estimates with race free equations

typically resulting in higher prevalence estimates in African-American populations and lower estimates in European populations when serum creatinine alone is used and smaller differences when cystatin C information is available.⁸⁸

The racial disparities in CKD are explained in part by the *APOL1* high-risk alleles, a genetic variant that when present in two copies confers approximately a twofold higher risk of CKD progression and ESKD.^{14,16,89,90} The gene variants provide resistance to infection with *Trypanosoma brucei rhodesiense*, a parasite endemic to Africa (Fig. 18.8).⁹¹ The exact mechanism for increased kidney risk is uncertain,⁹² and *APOL1* high-risk status—present in about 12% of African Americans but less than 1% of Whites in population-based studies in the United States⁹³—is thought to require a “second hit” to be pathogenic for kidneys. The presence of sickle cell trait, present in 7% of African Americans but virtually absent in Whites, also confers up to a twofold higher risk of ESKD, although less reliable across studies.^{18,19} Both *APOL1* high-risk status and sickle cell trait are present in greater proportions in the ESKD population than in the general population.^{94,95}

Other discovered genetic variants may also have effects on the development and progression of CKD, but they have generally had much smaller population prevalences or effect

sizes than *APOL1* risk alleles and the sickle cell trait.⁹⁶⁻⁹⁹ For example, genetic variation in *UMOD*, the gene coding for uromodulin, expressed in the thick ascending limb of the loop of Henle, may play a role in CKD. Uromodulin is the most commonly found protein in normal urine and likely affects renal salt handling.^{100,101} Common variants in the promoter region of *UMOD* have been associated with increased risk of CKD, CKD progression, ESKD, and hypertension in genome-wide association studies, albeit with a difference in the rate of eGFR decline in the range of 0.15 mL/min/1.73 m² per year.^{98,102-108} Genome-wide association studies have also identified genomic regions associated with more specific disease types, such as IgA nephropathy and idiopathic membranous nephropathy.^{24,109} Many genetic variants underlying monogenic disease have also been determined, such as variants in *PKD1* or *PKD2* causing autosomal dominant polycystic kidney disease and more than 20 genes involved in syndromic and nonsyndromic congenital anomalies of the kidney and urinary tract (a leading cause of kidney disease in children), but most of these genotypes are rare (<1 in 1000 for dominant diseases and <1 in 40,000 for recessive diseases).^{26,110,111}

Among clinical risk factors, the presence of diabetes mellitus and hypertension appear to be the most robustly associated with CKD. The presence of diabetes mellitus

was a risk factor for the development of CKD after full or partial nephrectomy¹¹² and also in several community-based studies.^{113,114} Diabetes has also been linked to the development of kidney failure and is considered the most common cause of ESKD in the world today. Evidence that tight glycemic control helps forestall the development of mild to moderate albuminuria in clinical trials further advances the notion that the presence of diabetes mellitus is on the causal pathway for the development and progression of many types of CKD.¹¹⁵ Interestingly, among people with CKD, diabetes is a weaker risk factor for ESKD once the GFR and albuminuria are accounted for since they represent mediators of the relation between diabetes and ESKD.⁷²

In contrast to the well-accepted role of diabetes in the development of CKD, there is ongoing debate as to whether hypertension is a cause of CKD, a consequence of CKD, or both. Many observational studies have linked higher blood pressure to the development and progression of CKD,¹¹⁶⁻¹¹⁸ but evidence from randomized controlled trials that lowering blood pressure helps forestall CKD progression is inconsistent.¹¹⁹⁻¹²¹ The Modification of Diet in Renal Disease study (*N* = 840) showed no benefit of the lower blood pressure goal on GFR decline during the trial period (mean, 2.2 years) but a beneficial effect on ESKD during longer follow-up. The African American Study of Kidney Disease and Hypertension (*N* = 1094) showed no benefit of the lower blood pressure goal on GFR decline over a mean of 4 years or on ESKD during longer follow-up.¹¹⁹ However, both studies suggested a larger benefit of the lower blood pressure goal in patients with higher proteinuria.^{122,123} The REIN-2 trial did not show a difference in progression to ESKD with a lower blood pressure goal, but this study achieved the lower blood pressure goal using a calcium channel blocker. This increases proteinuria, which may have blunted any beneficial effect.¹⁰⁶ In PKD, the HALT-PKD study showed a beneficial effect of a lower blood pressure goal on enlargement in total kidney volume but not on eGFR decline.¹²⁴ In SPRINT, a lower blood pressure goal appeared to result in a lower eGFR over the first 18 months, but eGFR trajectories remained relatively stable after that time, with a large beneficial effect of lower blood pressure on death and CVD, despite the higher risk of incident CKD.¹²⁵ The complexity of the relation of blood pressure treatment to CKD progression may not be surprising given the complex regulation of intrarenal hydraulic pressures, including by the afferent and efferent glomerular arterioles, and the central role of the kidney in regulating total blood volume, vascular resistance, and blood pressure.

Another increasingly prevalent risk factor for CKD is obesity. In a study of 320,252 participants in a U.S. health plan, obesity was associated with more than a threefold higher risk of developing ESKD over a mean of 26 years of follow-up. Contemporary studies with shorter follow-up have shown slightly more modest risks of CKD stages G4 to G5 associated with obesity, such as one in the United Kingdom, which estimated a twofold higher risk associated with body mass index (BMI) between 30 and 35 kg/m².¹²⁶ Similarly, a meta-analysis of ESKD risk among >4 million potential kidney donor candidates showed a 1.16 times higher risk per 5 kg/m² higher BMI over 30 kg/m².¹²⁷ The risk of CKD progression associated with obesity may be smaller still; in one cohort of Swedish people with CKD stages G4 to G5, BMI was unrelated to subsequent ESKD. On the other hand,

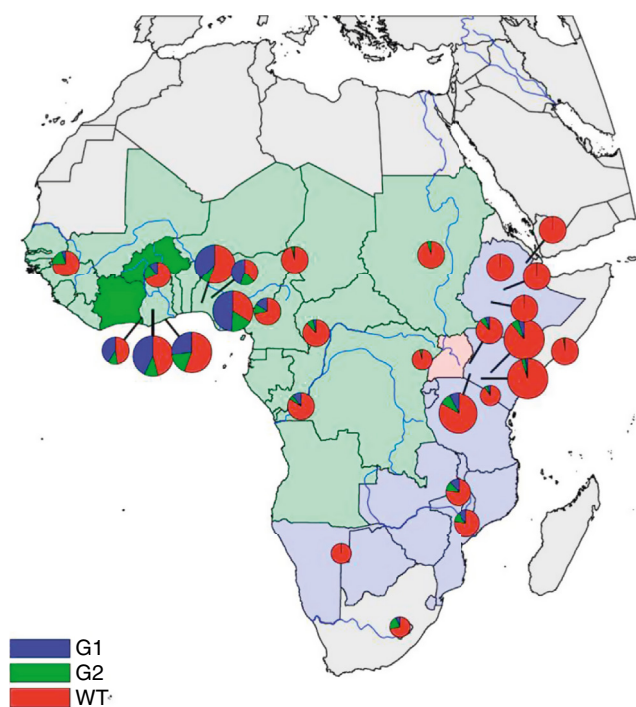


Fig. 18.8 Distribution of the G1 and G2 *APOL1* variants across Africa. Allele frequencies of the G1 and G2 variants are indicated as blue and green wedges, respectively. Circle size reflects the number of individuals genotyped: small, <10 individuals/20 chromosomes; medium, 10–100 individuals/20–200 chromosomes; large, >100 individuals/200 chromosomes. Countries are shaded according to the subspecies of *Trypanosoma brucei* that causes African sleeping sickness. Darker green, gambiense types 1 and 2; light green, gambiense type 1; pink, both rhodesiense and gambiense type 1; purple, rhodesiense. (From Thomson R, Genovese G, Canon C, et al. Evolution of the primate trypanolytic factor *APOL1*. *Proc Natl Acad Sci U S A*. 2014;111:E2130–E2139.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

weight loss achieved by lifestyle modification or bariatric surgery has been associated with a reduction in albuminuria and preservation of eGFR.¹²⁸⁻¹³⁰ For example, in Look AHEAD, a clinical trial in 5145 participants with diabetes, an intensive lifestyle intervention resulted in 4 kg of weight loss compared with the control arm and a 31% reduction in the development of CKD in very-high-risk categories, as defined by KDIGO. Obesity may contribute to CKD as a risk factor mediated by the development of diabetes and hypertension or with independent causal effects. In animal models, obesity results in glomerular hypertension and hyperfiltration, a proinflammatory milieu, and reduced adiponectin levels, all of which may contribute to CKD risk.¹³¹

There have been localized epidemics of kidney disease, such as Balkan nephropathy, those due to Chinese herbal supplements (now attributed to aristolochic acid¹³²), and the more recent Mesoamerican nephropathy,¹³³ which point to environmental factors that may additionally alter the risk of ESKD in populations.

END-STAGE KIDNEY DISEASE

METHODOLOGIC CONSIDERATIONS

In the United States, data regarding ESKD are generally excellent due to the establishment of comprehensive and well-maintained kidney disease registries by the National Institutes of Health in 1989. Today, the Annual Data Reports of the U.S. Renal Data System (USRDS) and the Scientific Registry of Transplant Recipients provide extensive epidemiologic information on incident and prevalent ESKD based on required reporting by providers of dialysis and transplantation services. As such, the epidemiology of ESKD reflects the clinical practice of initiation of KRT, as well as the progression of CKD to kidney failure. Information on untreated kidney failure—GFR < 15 mL/min/1.73 m² not treated by KRT—is not captured by the USRDS or Scientific Registry of Transplant Recipients. Some data sources have suggested that untreated kidney failure could represent a substantial number of kidney failure cases, particularly among the very old.^{52,74} Causes of kidney disease among those with ESKD are ascertained from the Centers for Medicare & Medicaid Services Form 2728 but are frequently missing and have not been validated.¹³⁴ Because dialysis is the

predominant mode of KRT, the epidemiology of ESKD largely reflects the epidemiology of dialysis. Specific topics related to the epidemiology of kidney transplantation are covered in a separate section.

PREVALENCE

As of December 31, 2020, there were 807,920 prevalent cases of ESKD in the United States or a rate of 2271/million population. Both figures decreased slightly over the preceding year, for the first time in 2 decades.³⁵ For reference, in 2000, the number of cases of ESKD was 389,592, or 1581 cases/million population (Fig. 18.9). Much of the growth in prevalence was attributable to the aging of the population (ESKD is more common in older populations) and growth in ESKD in those aged 75 years and older, in whom the sex- and race-adjusted prevalence of ESKD increased from 4639/million population in 2000 to 7172/million population in 2020. The growth in prevalence is also partly due to a decline in mortality, which decreased from 149/1000 patient-years to 133 deaths/1000 patient-years in 2019, although there was a large uptick presumably attributable to COVID in 2020.

In the United States, approximately 70% of prevalent ESKD cases were treated by dialysis, either hemodialysis (59.8%) or peritoneal dialysis (8.1%). The majority of ESKD cases were attributed to diabetes or hypertension, followed by glomerulonephritis and cystic kidney disease.

In 2022, the USRDS collected ESKD data from 49 different countries, enabling comparisons of both the absolute counts of ESKD cases and prevalence per million population. The prevalence of ESKD within the population varied substantially across the participating country, ranging from 109 per million population in Bangladesh to 3772 cases per million population in Taiwan. The Republic of Korea had the second highest prevalence, at 2708 per million population. All countries providing annual data since 2010 (*N* = 45) showed increases in prevalence (median, 19.6%). The increase was greatest in Thailand and the Republic of Korea, where the percent changes from 2010 to 2020 were 174% and 136%, respectively.

Large variations also existed among countries with respect to the distribution of KRT modality (Fig. 18.10). In Iceland, 71% of prevalent ESKD patients were treated

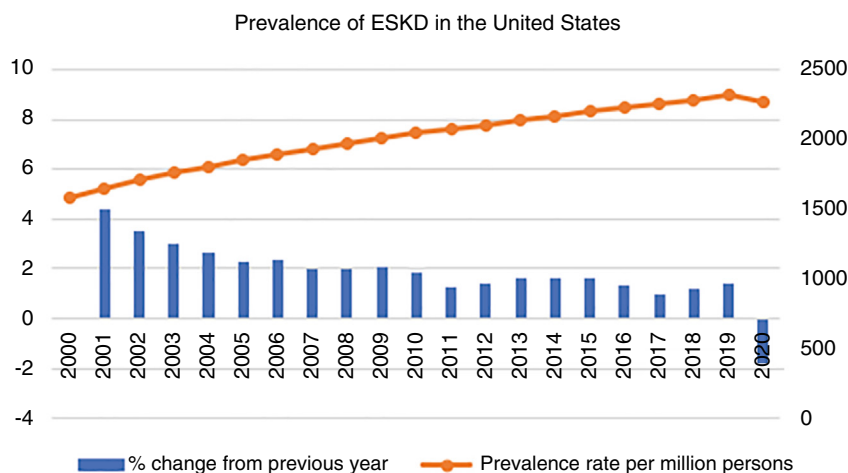


Fig. 18.9 Trends in adjusted ESKD prevalence (per million; trend lines; scale on left) and annual percentage (%) change in adjusted prevalence of ESKD (vertical lines; scale on right) in the U.S. population, 2000-2020, adjusted for age, sex, race, and ethnicity. The standard population was the U.S. population in 2015. (Data from U.S. Renal Data System. 2022 USRDS annual data report: epidemiology of kidney disease in the United States.)

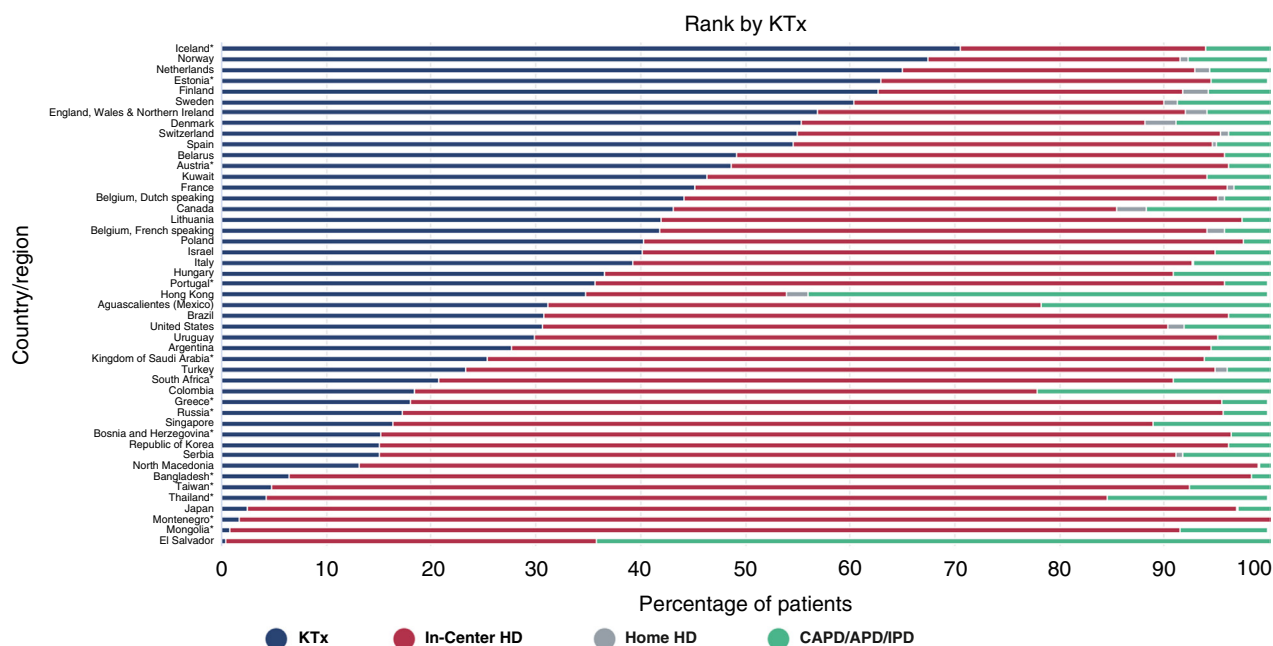


Fig. 18.10 Percentage distribution of type of kidney replacement therapy modality used by patients with end-stage kidney disease. By country, in 2020. HD, Hemodialysis. (From U.S. Renal Data System. 2022 USRDS annual data report: epidemiology of kidney disease in the United States.)

with a functioning transplant, 24% received in-center hemodialysis, and 6% received peritoneal dialysis in 2020. Peritoneal dialysis was most common in El Salvador, where 64% received peritoneal dialysis and 35% received in-center hemodialysis. In contrast, in-center hemodialysis comprises $\geq 95\%$ of the ESKD in Japan and Montenegro. Additional information on differences in hemodialysis practice by country was collected by the Dialysis Outcomes and Practice Patterns Study between June 1996 and March 2012.¹³⁵ Interestingly, in the population of 35,964 hemodialysis patients sampled from 12 countries, the proportion of women was much lower than that of the general population. This appeared attributable to a greater incidence of ESKD among men rather than better survival because the male-to-female mortality ratio was close to 1.

Evidence suggests that the variation in ESKD prevalence has much to do with the resources available to provide KRT. One study has suggested that the gap between people worldwide who require KRT and those who undergo KRT was at least 2.3 million in 2010.⁷⁷ The authors estimated the prevalence of ESKD undergoing KRT to be 1839 cases/million people in North America, followed by 719 cases/million people in Europe, 686 cases/million in Oceania, and 626 cases/million in Latin America. Asia, which was reported to have an ESKD prevalence of 232 cases/million population, was estimated to have the highest absolute number of people with ESKD, at nearly 1 million cases in 2010. Gross national product was highly associated with prevalence, with the highest-income countries reporting a much higher prevalence of patients undergoing KRT. Interestingly, this relationship was much stronger than the one between the prevalence of diabetes and prevalence of patients undergoing KRT (Fig. 18.11). The International Society of Nephrology Global Kidney Health Atlas provides a

detailed summary of the variation in access to KRT, ranging from 40% in Africa in 2023 to 100% in Europe, the Middle East, and North and East Asia.^{12,136}

INCIDENCE

The incidence, or rate of new cases, of ESKD is also reported annually by the USRDS. In the United States, the number of incident cases of ESKD in 2020 was 130,522, or an incidence rate of 393 per million per year. These figures are 3.3% and 6.2% decreases, respectively, over the prior year. However, since 2000, the trend has been a relatively steady increase: The number of incident cases of ESKD was 94,466, or 330 cases per million population. Much of this increase in incidence rates can be attributed to aging of the U.S. population.

Most of the new ESKD cases initiated KRT using hemodialysis (83.9%), whereas 12.7% used peritoneal dialysis and 3.1% received a kidney transplant without prior dialysis. These proportions have remained relatively stable over the past decade. The adjusted incidence rate for Black/African American individuals was 949 cases per million persons in 2020, down from 1135 per million persons in 2000. Declines were also seen in the Native American population (893 to 596 cases per million persons) and U.S. Hispanic population (695 to 511 cases per million persons). In contrast, the adjusted incidence of ESKD within Asian and White populations was relatively flat over the past 2 decades (371 to 349 cases per million persons and 270 to 249 cases per million persons, respectively).

When extended to estimate the lifetime incidence of ESKD, racial and ethnic disparities remain.^{49,137,138} One study using 2013 USRDS data has suggested that the lifetime incidence of ESKD is 3.1% for a non-Hispanic white man,

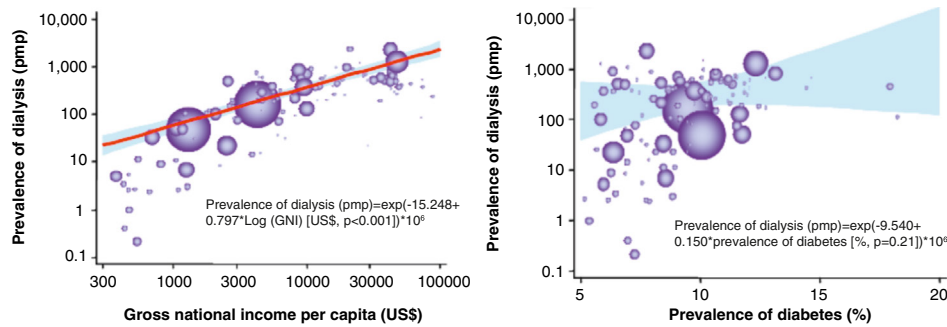


Fig. 18.11 Association between prevalence of end-stage kidney disease (ESKD) treated by dialysis in 123 countries (gross national income per capita [left], prevalence of diabetes [right]). In both graphs, China and India are represented by the largest and second largest circles, respectively. (From Liyanage T, Ninomiya T, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet*. 2015;385:1975–1982.)

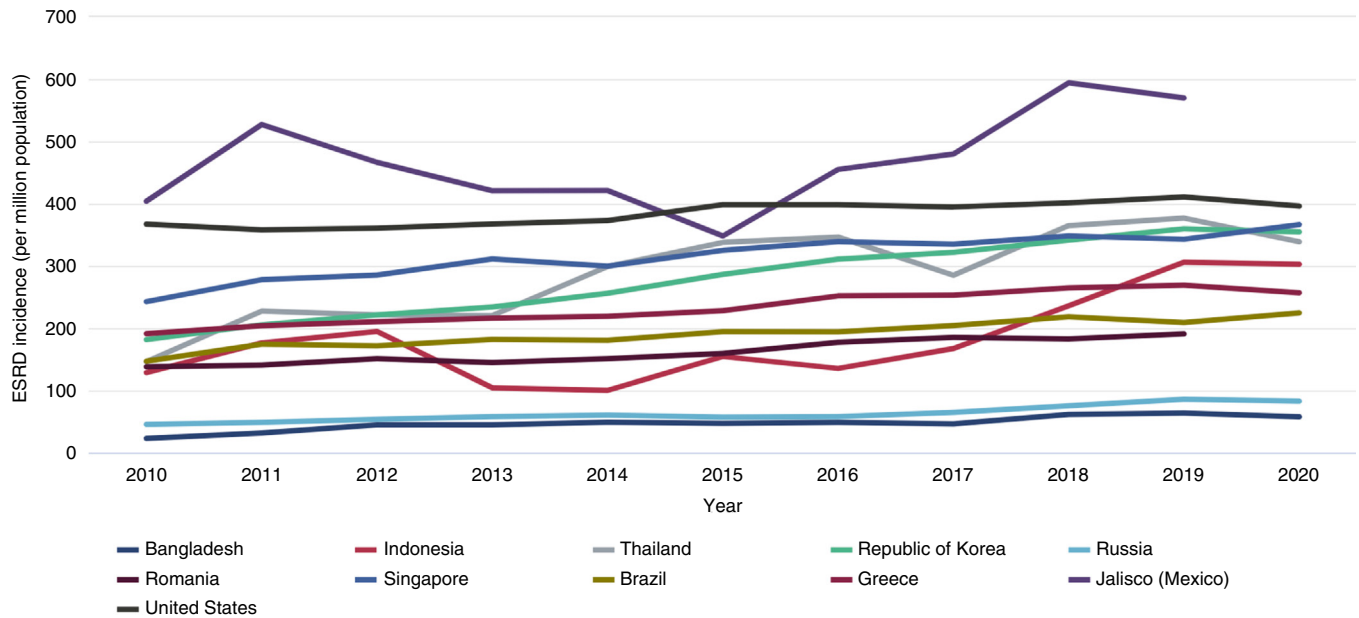


Fig. 18.12 Trends in the incidence rate of treated end-stage kidney disease (ESKD; per million population/year), by country, 2010–2020. The 10 countries with the highest percentage rise in ESKD incidence rate over the past decade, plus the United States. (From U.S. Renal Data System. 2022 USRDS Annual data report: epidemiology of kidney disease in the United States.)

6.2% for a Hispanic man, and 8.0% for a non-Hispanic Black man.¹³⁷ Furthermore, these disparities changed little over the previous decade.

When compared with other countries, the incidence of ESKD in the United States was among the highest, trailing only Taiwan in 2020. Other high ESKD incidence countries included Singapore, the Republic of Korea, Thailand, Japan, and Indonesia. Among the represented countries, the lowest incidence rate per million population/year was in South Africa with 11 ESKD cases per million per year. Trends in the incidence of ESKD varied by country, with many of the Northern European countries experiencing lower ESKD incidence over time. In contrast, there were large increases in incidence rates in Thailand, the Republic of Korea, Indonesia, and Mexico (Fig. 18.12).

Interestingly, the proportion of new cases attributed to diabetes also varied by country. In Asia, Latin America, and the United States, many countries reported >40% of the new cases as attributable to diabetes. In contrast, many European countries reported fewer incident cases attributable to diabetes (Fig. 18.13). In addition, in every country except Kuwait, there was a disparity in new ESKD by sex. Men had

higher rates of new ESKD than women, sometimes even twofold higher.

OUTCOMES

Age-adjusted mortality in ESKD is much higher than in the general population.^{139,140} In 2020, the USRDS reported that the adjusted mortality rate for people with ESKD was 152 deaths per 1000 patient-years. For patients receiving dialysis, adjusted mortality rates were 186 deaths per 1000 patient-years.³⁵ In the year after dialysis initiation, the risk of death peaked at 2 months for patients receiving hemodialysis and increased slowly over the year for patients receiving peritoneal dialysis. Approximately 41% of all deaths among dialysis patients in 2012 were attributed to CVD.

Mortality rates in incident dialysis patients vary by age and underlying comorbid conditions. Early mortality is more common among people with lower serum albumin and phosphorus levels, as well as those with congestive heart failure and cancer.¹⁴¹ Although racial differences in dialysis mortality have been noted, particularly in older age groups, they may be due in part to differences in the rate

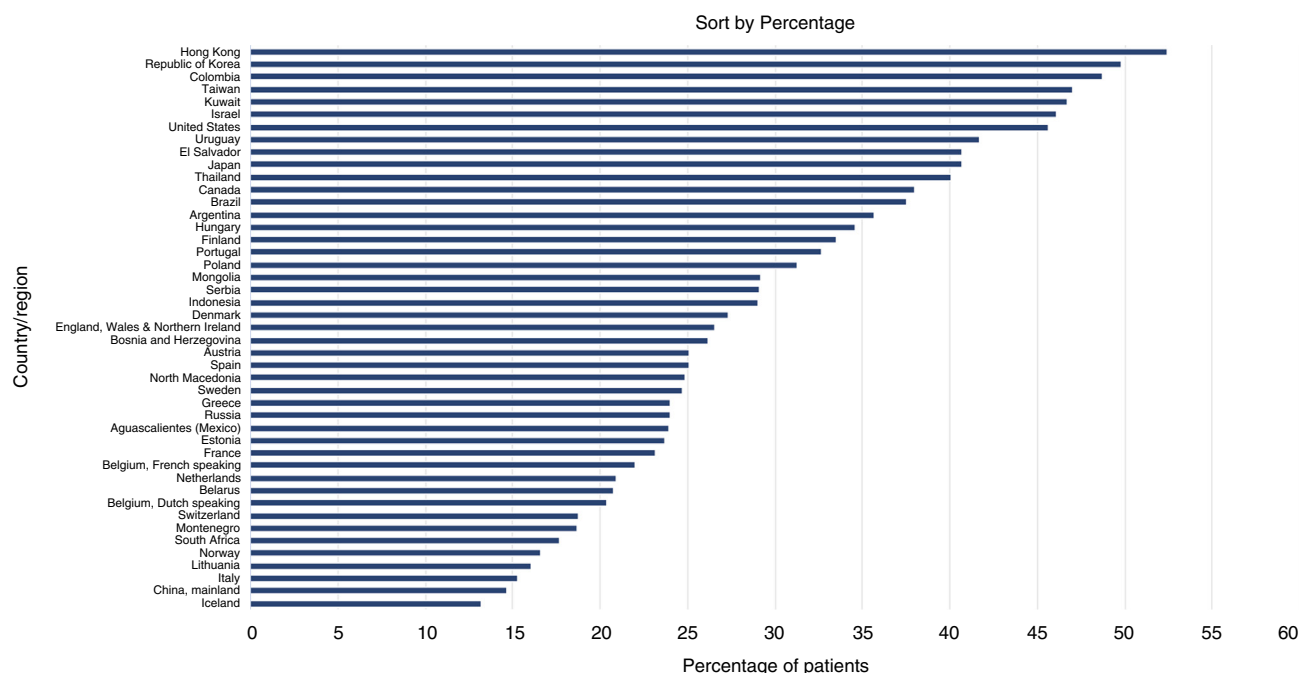


Fig. 18.13 Proportions of incident end-stage kidney disease (ESKD) due to diabetes, by country, 2020. (From U.S. Renal Data System. 2022 USRDS annual data report: epidemiology of kidney disease in the United States.)

of transplantation, the level of residual kidney function at dialysis initiation, and the presence and severity of comorbid conditions.¹⁴²

Mortality rates in incident dialysis patients also vary by country. In patients selected for the Dialysis Outcomes and Practice Patterns Study, the mortality rate ranged from 17 deaths/100 patient-years in the first 120 days in Japan to 33.5 deaths/100 patient-years in the first 120 days in Belgium.¹³⁹ Long-term mortality was even more disparate: After the first year on dialysis, there were 5.2 deaths/100 patient-years in Japan compared with 19.9 deaths/100 patient-years in Belgium. It should be noted that the proportion of patients 75 years of age and older was smaller in Japan; however, adjusted mortality rates were also lower in Japan than in any other included countries. In general, European and North American long-term mortality rates were fairly similar, with the United States reporting the highest adjusted mortality rates (Fig. 18.14).

It should also be noted that mortality on dialysis is sometimes related to withdrawal of therapy. In Australia and New Zealand, nearly 40% of deaths within the first 120 days were coded as being due to withdrawal from care, compared with much lower rates in regions such as Italy, France, and Germany.¹³⁹

Trends in mortality have suggested that dialysis patients in the United States are living longer compared with previous decades, with the overall mortality rate decreasing by 32% since 1996. Although there have been numerous improvements in dialysis treatment, their relative contribution to the trend toward lower mortality is uncertain. With the exception of antihypertensive treatment, controlled trials evaluating single interventions have not shown an effect on all-cause mortality in dialysis patients.¹⁴³⁻¹⁵¹ In our view, this may be due in part to the difficulty of showing a reduction in all-cause mortality with

interventions targeting a single risk factor. The adoption of multiple interventions in practice may achieve this goal.

RISK FACTORS

Risk factors for the development of ESKD demonstrate some notable differences from those that portend a higher risk of incidence of earlier stages of CKD. Most notably, age has often demonstrated an inverse association with ESKD risk, adjusted for other factors.^{73,152,153} Given the strong association between older age and incidence of CKD, the reverse association is somewhat surprising. Possible explanations are that the competing event of mortality becomes much more common in older adults, and many older adults may choose not to initiate dialysis. Indeed, in Canada, people who were 85 years or older with an eGFR from 15 to 29 mL/min/1.73 m² experienced 20 cases of untreated kidney failure/1000 patient-years and 1.5 cases/1000 patient-years of treated kidney failure (i.e., undergoing dialysis or transplantation).⁷⁴ In contrast, for people ages 18 to 44 years with a similar eGFR, the rates were 3.5 cases/1000 person-years and 17 cases/1000 person-years of untreated and treated kidney failure, respectively. Similar observations have been demonstrated in the United States.⁵² Finally, it is possible that eGFR decline is slower among older compared with younger individuals.¹⁵⁴

Sex and race are additional risk factors that display somewhat disparate associations with the incidence of earlier stages of CKD compared with ESKD. For example, although women may be at a slightly higher risk for eGFR <60 mL/min/1.73 m²,¹⁵⁵ men make up a greater proportion of incident and prevalent ESKD patients. The higher risk of ESKD among men persists, even after adjusting

for demographic and clinical factors. Reasons for ESKD differences by sex may stem from hormonal differences¹⁵⁶ or simply from differences in care delivery and preference. For example, women tend to start dialysis at a lower eGFR than men.¹⁵⁷ In contrast, despite later initiation of dialysis among African Americans, African Americans have nearly a fourfold higher ESKD risk than Caucasians.^{28,49,157} These disparities persist after adjustment for socioeconomic status,

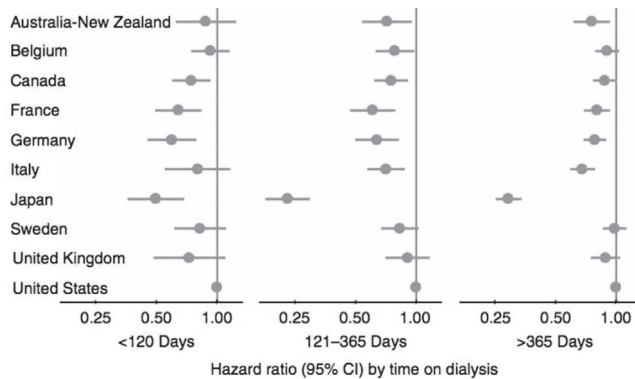


Fig. 18.14 Mortality in each DOPPS (Dialysis Outcomes and Practice Patterns Study) country versus the United States by time on hemodialysis (HD). Model adjusted for age, sex, race, and diabetes as the cause of end-stage kidney disease, stratified by study phase ($N = 86,886$ HD patients from DOPPS census [2002–2008]). CI, Confidence interval. (From Robinson BM, Zhang J, Morgenstern H, et al. Worldwide, mortality risk is high soon after initiation of hemodialysis. *Kidney Int.* 2014;85:158–165.)

clinical characteristics, and access to health care.^{84,158,159}

As discussed previously, diabetes mellitus is considered to be the major cause of ESKD worldwide, in part due its growing incidence and prevalence.^{28,160} In Singapore, 68% of incident ESKD cases in 2020 were attributed to diabetes; the corresponding figure in the United States was 46%. In a USRDS survey of 45 world regions, only 8 reported that <20% of ESKD cases were due to diabetes (Switzerland, Montenegro, South Africa, Norway, Lithuania, Italy, mainland China, and Iceland).³⁵ Furthermore, diabetes mellitus can be considered as a risk factor for both CKD incidence and disease progression. Diabetic CKD is frequently characterized by the presence of albuminuria, which is also a risk factor for CKD progression to ESKD.^{71,72} The second most common listed cause for ESKD in the United States is hypertension, although it is believed that many of the cases ascribed to hypertension may be due to other factors, such as *APOL1* risk alleles or parenchymal disease. Obesity has also variably been associated with increased ESKD risk, with older studies with longer follow-up demonstrating stronger associations than more contemporary studies.^{127,161}

Geographic region accounts for some of the differences in ESKD risk as well. A systematic review reported differences in ESKD prevalence where North America had the highest prevalence, at 1839 cases/million people, followed by Europe, at 719 cases/million people.⁷⁷ These differences appeared driven in large part by income of the country, with an estimated 74-fold difference in prevalence between high-income and low-income countries.

KIDNEY TRANSPLANTATION

PREVALENCE

In the United States, approximately 31% of prevalent ESKD cases have a functioning kidney transplant, corresponding to 245,846 people in 2020. The number of prevalent transplants has more than doubled over the past 2 decades, from 107,109 in 2000. The prevalence is currently one of the highest in the world, at 746 people with a functioning kidney transplant per million population in the United States, but other countries such as Spain (744 cases per million) and Portugal (716 cases per million) also have high prevalence. Prevalence in all assessed countries except Saudi Arabia, Qatar, and Kuwait increased over the past decade. As a proportion of prevalent ESKD cases, the highest proportions treated with kidney transplant were generally found in Northern Europe, where >50% of ESKD patients were living with a functional kidney transplant.

INCIDENCE

In the United States, there were 25,487 kidney transplants in 2021, the highest number to date (Fig. 18.15).¹⁶² There were 5949 living donor transplants. Waiting time was significantly different by blood type, PRA, and region. Using 2020 data, kidney transplant rates per 1000 patient-years also varied: highest among people aged 17 years and younger (391) and lowest among those aged 75 years and older (4). On average, women had slightly lower transplant rates than men (35 vs. 40 per 1000 dialysis patient-years) as did Black patients compared with other races (30 vs. 42, 33, and 42 for White, Hispanic, and Asian dialysis patients, respectively).

Internationally, there is even greater variation in kidney transplantation rates due in part to differences in health care systems, availability of organs, and cultural beliefs. In 2014, the highest transplantation rates/1000 dialysis patients were seen in Northern Europe, with Norway and Finland the highest at 135 and 133 transplants per 1000 dialysis patients.³⁵

OUTCOMES

Randomized controlled trials have not been performed, but kidney transplantation is generally accepted as the preferred KRT modality, with lower mortality and better quality of life compared with hemodialysis or peritoneal dialysis.¹⁶³ These benefits are thought to extend to recipients of both living and deceased donor organs including those who receive organs with so-called marginal characteristics.^{164–166} Benefits of transplantation are also seen in older recipients; for example, patients older than 70 years who received transplants had a 41% lower risk of death compared with wait-listed counterparts in adjusted analysis. Whereas cardiovascular mortality increases with dialysis vintage, there is no such increase in cardiovascular events with kidney transplantation vintage, suggesting that some of the mortality benefits seen with transplantation may be a reduction in the development of CVD.¹⁶⁷

On the other hand, kidney transplantation is not without risk. The leading cause of death is CVD. Among risk factors

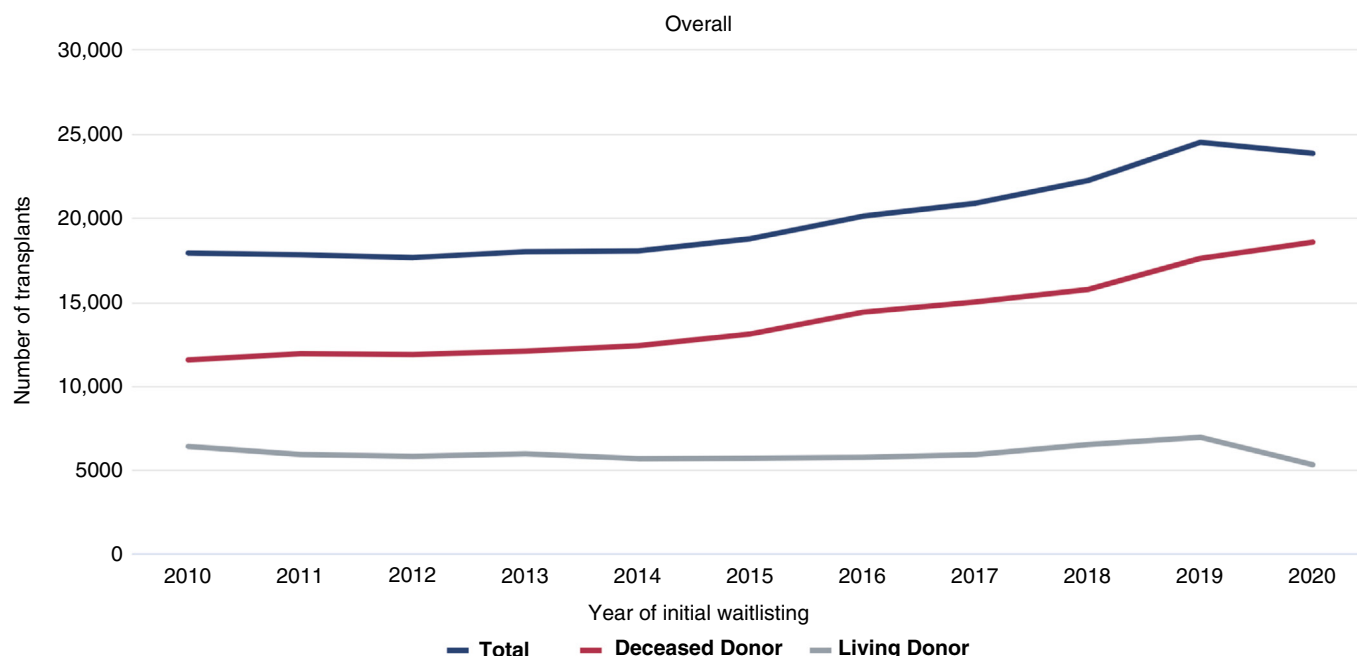


Fig. 18.15 Number of kidney transplants by donor type, 2010-2020, in patients undergoing dialysis. (Data from U.S. Renal Data System. 2022 Annual data report: atlas of chronic kidney disease and end-stage renal disease in the United States.)

for CVD, new-onset diabetes mellitus after transplantation is of particular relevance because it may be increased by the calcineurin inhibitor tacrolimus, a commonly used agent for immunosuppression. New-onset diabetes is variably captured but has been reported in up to 50% of transplant recipients, with a higher risk among older recipients.¹⁶⁸ Infections are the second leading cause of death in transplant recipients.^{169,170} Cancer is the third leading cause of death, with the incidence of cancer mortality twofold to threefold higher than the general population.^{171,172}

DISPARITIES IN KIDNEY TRANSPLANTATION

Several international studies have suggested disparities in rates of kidney transplantation by socioeconomic status and race, particularly for living donor kidney transplants.^{173,174} In the United States, disparities have been documented in transplant referral, recipient candidate approval, placement on the waiting list, time on the waiting list, and allograft survival.¹⁷⁴⁻¹⁷⁹ It has been estimated that eligible African American transplant candidates receive 35% fewer transplants than eligible Caucasian transplant candidates.¹⁸⁰ A study in Australia has found that recipients from socioeconomically advantaged areas were more likely to receive a living kidney donor transplant than those of lower socioeconomic status, although rates of deceased donor transplantation were similar. Some of the disparities in living donor transplantation may be due to increased risk of CKD in family members due to the presence of genetic risk variants. Similarly, the presence of the *APOL1* high-risk genotype in the allograft (but, interestingly, not the recipient) is a risk factor for shortened allograft survival.¹⁸¹

ACUTE KIDNEY INJURY

METHODOLOGIC CONSIDERATIONS

There are many similarities in the epidemiology of AKI and CKD, in part because the definitions for both include a criterion based on a measure of GFR (see Table 18.1). The urine output criterion is unique to the definition of AKI, but it is also not captured in much of the previous literature and has been criticized for both logistic and theoretic reasons.^{182,183} Studies of AKI typically capture disease by changes in creatinine level (with a wide range of definitions) or by relying on billing codes, such as those classified in the *International Classification of Disease, Ninth Clinical Modification*.¹⁸² Studies using diagnostic codes to identify AKI events can be problematic since labeling an event as AKI depends on the provider's perception of what constitutes AKI. Using a creatinine-based definition of AKI, the sensitivity of diagnostic codes has been shown to vary by era and patient age.¹⁸⁴

INCIDENCE

By definition, AKI is a transient disease; kidney disease lasting longer than 3 months is defined as CKD.^{4,7} Thus AKI can be described only through incidence rather than prevalence. One study of community-dwelling people in Northern California has estimated the incidence rate to be 522.4 cases of AKI/100,000 person-years and 29.5 cases of dialysis-requiring AKI/100,000 person-years. A population-based study from Scotland has estimated the AKI and AKI requiring dialysis rates as 214.7 and 18.3 cases/100,000 patient-years, respectively. In a

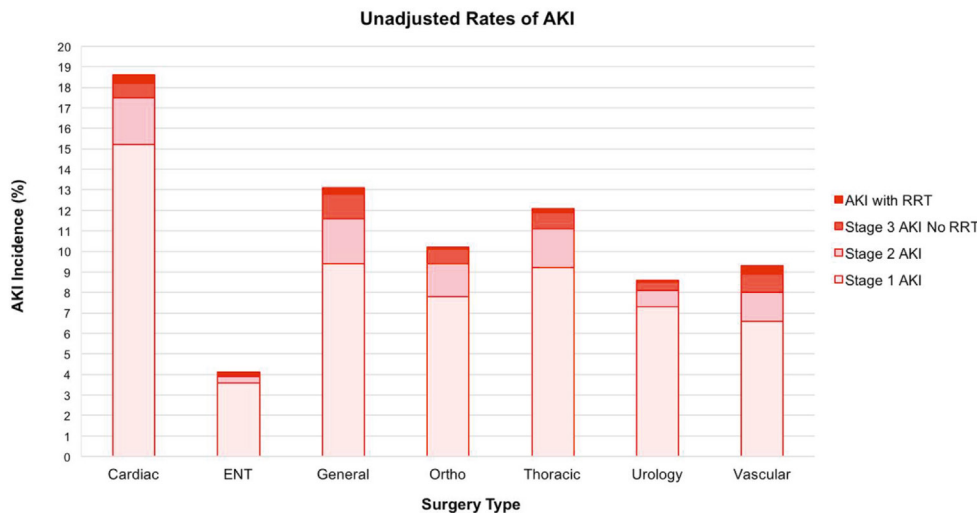


Fig. 18.16 Unadjusted rates of acute kidney injury, by surgery type and stage. AKI, Acute kidney injury; ENT, ear, nose, throat; RRT, renal replacement therapy. (Adapted from Grams ME, Sang Y, Coresh J, et al. Acute kidney injury after major surgery: a retrospective analysis of Veterans Health Administration data. *Am J Kidney Dis.* 2016;67[6]:872–880.)

Canadian cohort of patients with both creatinine and urine protein measurements, incidence rates were much higher, ranging from 100 to 11,700 cases of hospitalized AKI/100,000 person-years, depending on the level of GFR and proteinuria.¹⁸⁵ In patients with Medicare insurance and older than 65 years, 6.2% had a hospitalization with AKI over the course of 1 year, as identified by a billing code for AKI.

In a hospitalized population, the rate of AKI may be even higher. A meta-analysis of hospitalized patients worldwide has estimated that almost 25% had AKI ascertained using the KDIGO definition; approximately 10% of cases required RRT.¹⁸⁶ A similar estimate was derived from U.S. Department of Veterans Affairs patients and, notably, only 49% of those with AKI ascertained by the serum creatinine criterion had an associated AKI billing code. Rates of AKI have also been defined in certain hospital settings, such as after major surgery and during hospitalizations with coronavirus disease 2019 (COVID-19). Estimates using Veterans Affairs data have suggested that the 7-day incidence of AKI after major surgery ranges from 4.1% after ear, nose, and throat surgery to 18.7% after cardiac surgery (Fig. 18.16).¹⁸⁷ U.S. and European estimates of patients hospitalized with COVID-19 suggest that approximately one-quarter developed AKI, with up to 20% requiring RRT when requiring intensive care unit admission.^{188–191} COVID-19 complicated by AKI was associated with a significantly higher mortality than without AKI (60.5% vs. 27.4%).¹⁹² Causes of AKI are rarely assessed in large studies, but some studies have suggested that among hospitalized patients, approximately two-thirds of AKI episodes were community acquired and one-third were hospital acquired.¹⁹³

OUTCOMES

The study of AKI has been prioritized in both clinical and basic science research because of the recognition that outcomes, including development and progression of CKD, are worse among patients who experience even a small increase in serum creatinine level compared with those with stable values. In addition, studies have suggested that oliguria, even when not accompanied by a rise in serum

creatinine level sufficient to meet the AKI criterion, may confer higher risk.^{184,194} The incidence of CKD, decline in eGFR, ESKD, and death are all higher in patients who experience AKI, with higher AKI stage conferring higher risk.^{182,195–199} For example, in one U.S. study, the risk of developing stage G4 or G5 CKD was 28 times higher after surviving an episode of KRT-requiring AKI.²⁰⁰ Even modest changes in serum creatinine level have been associated with higher costs and length of stay. One older study estimated that an increase in serum creatinine level of 0.5 mg/dL or higher was associated with 3.5 additional hospital days and \$7500 in excess costs. The absolute risk of adverse outcomes varies based on underlying characteristics and stage of AKI, but many of those with AKI severe enough to require KRT never recover kidney function fully.^{196,200,201} Controversial is whether the adverse outcomes are caused by AKI or simply due to shared risk factors.^{202,203} No trials have definitively proven causation due to the lack of effective therapy for AKI prevention and the need for large sample sizes and longer-term follow-up.²⁰⁴

RISK FACTORS

Risk factors for AKI are similar to those for CKD: older age, male sex, and variably, Black race.^{35,58} Racial disparities in AKI may be confounded by differences in socioeconomic status—itsself a risk factor for AKI—and, in a small study, were not explained by differences in kidney risk variants in APOL1.²⁰⁵ Diabetes also confers an increased risk for AKI, with odds of AKI from 1.5- to 2.5-fold higher among persons with diabetes.^{59,206} Heart failure is also strongly associated with AKI.³⁵ The strongest risk factors, however, appear to be lower eGFR and higher albuminuria, with each conferring a higher risk in a graded fashion and independent of the risks associated with diabetes mellitus or hypertension (Fig. 18.17).³⁵

EVIDENCE GAPS AND LIMITATIONS OF THE LITERATURE

The literature of the epidemiology of kidney disease has important gaps:

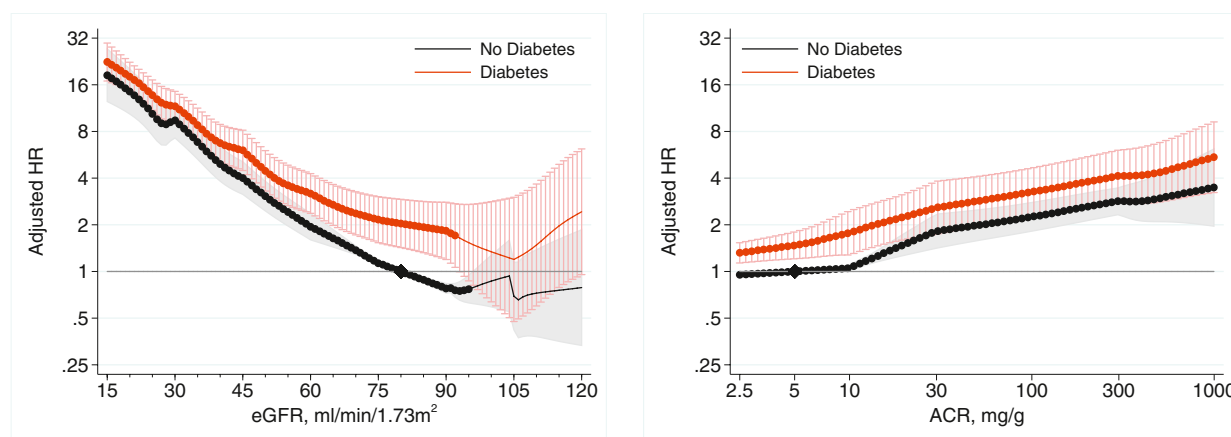


Fig. 18.17 Acute kidney injury risk by estimated glomerular filtration rate (eGFR) and urine albumin-to-creatinine ratio (ACR), stratified by diabetes status. Hazard ratios (HRs; with 95% CI [confidence interval]) of acute kidney injury, using common referent group without diabetes, according to eGFR (reference, 80 mL/min/1.73 m² without diabetes; left) and ACR (reference, 5 mg/g without diabetes; right) in individuals with and without diabetes in general population cohorts. (From James MT, Grams ME, Woodward M, et al. A meta-analysis of the association of estimated GFR, albuminuria, diabetes mellitus, and hypertension with acute kidney injury. *Am J Kidney Dis.* 2015;66:602–612.)

1. Studies of the onset of disease are limited by the timing of (and indication for) ascertainment of kidney disease measures since kidney disease is often asymptomatic. Serum creatinine level is frequently measured during health encounters for acute and chronic illness. Cystatin C and markers of kidney damage such as albuminuria are measured less frequently than serum creatinine level in clinical practice, and thus AKD and CKD are often underestimated. In research cohorts, measurements of serum creatinine levels are generally not frequent enough to distinguish AKI, AKD, and CKD.
2. Factors other than GFR can affect serum concentrations of creatinine and cystatin C, and correlation of the non-GFR determinants of these filtration markers with measured and unmeasured characteristics can distort associations.
3. Many identified risk factors do not meet all the criteria for causality outlined by Bradford Hill.¹⁰ The regulatory roles of the kidney and possible bidirectional causality complicate inferences from observed associations.
4. Definitions for cause of kidney disease are less well established than that of G- and A-staging. The cause of disease is often multifactorial, and the clinical information necessary for ascertaining cause of disease is generally not available in research studies and clinical populations.
5. ESKD is defined by a therapy that is not uniform in its implementation. Symptoms of kidney failure are poorly defined and ascertained, and the availability of KRT varies widely among countries, complicating inferences about kidney failure from ESKD studies.
6. Laboratory methods for the assessment of kidney measures can differ. Guidelines have improved comparability across studies by recommending standardization of assays for serum creatinine and cystatin C levels and routine reporting of eGFR, as well as measurement of urine albumin rather than total protein, but additional work is needed.

CONCLUSION

In conclusion, kidney disease is now recognized as a pressing global health concern. More than 10% of the world's population has CKD, and nearly 25% of the global hospitalized population experiences AKI. Enormous disparities exist in access to treatment, and millions of people are undertreated or untreated. Kidney disease epidemiology plays a critical role in providing evidence for the issuance of guideline definitions and recommendations, harmonization of research, development of trial endpoints, and success of global initiatives to improve the health of patients who suffer from kidney disease of all stages.

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QUESTIONS

1. A 78-year-old White man with diabetes, hypertension, and heart failure with preserved ejection fraction presents with a serum creatinine level of 1.2 mg/dL, eGFRcr of 58 mL/min/1.73 m², and a spot urine albumin-to-creatinine ratio of 1000 mg/g. According to the Kidney Disease Improving Global Outcomes (KDIGO) chronic kidney disease (CKD) staging and classification, he has:
 - a. No CKD, moderate risk
 - b. CKD stage G3A2, moderate risk
 - c. CKD stage G2A3, moderate risk
 - d. CKD stage G3A3, very high risk
3. Using the kidney failure risk equation, for the same age, sex, glomerular filtration rate, and level of albuminuria, a person in Europe would be predicted to have a _____ risk of end-stage kidney disease compared with a person in Canada.
 - a. Higher
 - b. Lower
 - c. The same

Answer: b

- Answer: d**
2. A 24-year-old football player presents with a serum creatinine level of 1.7 mg/dL and eGFRcr of 55 mL/min/1.73 m². What additional test(s) might you order next to evaluate better whether he has kidney disease and to determine his prognosis?
 - a. Spot urine albumin and creatinine
 - b. Serum cystatin C
 - c. Serum protein electrophoresis
 - d. a and b
 - e. All of the above
 4. Which country has the highest prevalence of ESKD treated with dialysis?
 - a. United States
 - b. Thailand
 - c. Taiwan
 - d. Japan
 - e. Belgium

Answer: c

Answer: d